



## Qualification Pack

# Quality Technician - Construction

QP Code: ICE/CON/Q5001

Version: 1.0

NSQF Level: 4

Integrated Council for Entrepreneurship and Skilling || 533-R  
Model Town Ludhiana || email:cmswami@ice.net.in



## Qualification Pack

## Contents

|                                                                                                                           |    |
|---------------------------------------------------------------------------------------------------------------------------|----|
| ICE/CON/Q5001: Quality Technician - Construction .....                                                                    | 3  |
| <i>Brief Job Description</i> .....                                                                                        | 3  |
| Applicable National Occupational Standards (NOS) .....                                                                    | 3  |
| <i>Compulsory NOS</i> .....                                                                                               | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                                                           | 4  |
| ICE/CON/N5001: Conduct construction material testing using the Universal Testing Machine (UTM) ...                        | 6  |
| ICE/CON/N5002: Conduct Ground Penetrating Radar (GPR) testing .....                                                       | 12 |
| ICE/CON/N5003: Carry out Digital Concrete Testing .....                                                                   | 18 |
| ICE/CON/N5004: Carry out Ultrasonic Testing (UT) of construction materials .....                                          | 24 |
| ICE/CON/N5005: Carry out Eddy Current Testing (ECT) of construction materials .....                                       | 30 |
| ICE/CON/N5006: Conduct tests on soil, aggregate, bitumen, steel, different construction mixes, and NDT on concrete .....  | 36 |
| ICE/CON/N5007: Carry out Highway Testing through FWD, NSV, Benkelman Beam, and Total Station .....                        | 44 |
| ICE/CON/N5008: Set up the project civil lab .....                                                                         | 51 |
| ICE/CON/N5009: Basics of plants and machinery (Hot Mix Plant, Batching Plant, Paver, Grader, Loader) .....                | 58 |
| ICE/CON/N5010: Basics of preparation of lab program and reporting daily progress report and monthly progress report ..... | 66 |
| ICE/CON/N5011: Follow health, safety, and environmental standards in construction material testing .....                  | 72 |
| DGT/VSQ/N0102: Employability Skills (60 Hours) .....                                                                      | 79 |
| Assessment Guidelines and Weightage .....                                                                                 | 86 |
| <i>Assessment Guidelines</i> .....                                                                                        | 86 |
| <i>Assessment Weightage</i> .....                                                                                         | 87 |
| Acronyms .....                                                                                                            | 89 |
| Glossary .....                                                                                                            | 90 |



## Qualification Pack

### ICE/CON/Q5001: Quality Technician - Construction

#### Brief Job Description

A Quality Technician - Construction is responsible for conducting tests on materials used in highway and infrastructure projects, including soil, aggregate, bitumen, steel, and concrete mixes like GSB, WMM, DBM, and BC. The role involves using equipment such as the Universal Testing Machine (UTM) for strength assessments and conducting non-destructive testing (NDT) on concrete. The technician also performs specialized tests like FWD and NSV and is involved in setting up project labs, including lab layout, equipment foundations, and testing platforms. Additionally, the technician prepares lab programs and reports, including DPR and MPR.

#### Personal Attributes

The individual should be physically fit to work in varied conditions for long duration. The person should have attention to detail, technical proficiency, analytical thinking and problem solving skills. The individual should be adaptable and safety conscious with good verbal and written communication skills.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [ICE/CON/N5001: Conduct construction material testing using the Universal Testing Machine \(UTM\)](#)
2. [ICE/CON/N5002: Conduct Ground Penetrating Radar \(GPR\) testing](#)
3. [ICE/CON/N5003: Carry out Digital Concrete Testing](#)
4. [ICE/CON/N5004: Carry out Ultrasonic Testing \(UT\) of construction materials](#)
5. [ICE/CON/N5005: Carry out Eddy Current Testing \(ECT\) of construction materials](#)
6. [ICE/CON/N5006: Conduct tests on soil, aggregate, bitumen, steel, different construction mixes, and NDT on concrete](#)
7. [ICE/CON/N5007: Carry out Highway Testing through FWD, NSV, Benkelman Beam, and Total Station](#)
8. [ICE/CON/N5008: Set up the project civil lab](#)
9. [ICE/CON/N5009: Basics of plants and machinery \(Hot Mix Plant, Batching Plant, Paver, Grader, Loader\)](#)
10. [ICE/CON/N5010: Basics of preparation of lab program and reporting daily progress report and monthly progress report](#)
11. [ICE/CON/N5011: Follow health, safety, and environmental standards in construction material](#)



## Qualification Pack

[testing](#)

12. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

### Qualification Pack (QP) Parameters

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                             | Construction                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Sub-Sector</b>                                         | Real Estate and Infrastructure Construction                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Occupation</b>                                         | Quality Assurance and Quality Control                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>NSQF Level</b>                                         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Credits</b>                                            | 18                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/3112.9900                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minimum Educational Qualification &amp; Experience</b> | 12th grade Pass (Physics Chemistry Mathematics) with NA of experience<br>OR<br>11th grade pass with 1.5 years of experience relevant experience in roads and highways construction<br>OR<br>10th grade pass with 3 Years of experience relevant experience in roads and highways construction<br>OR<br>Previous relevant Qualification of NSQF Level (3.5 level) with 1.5 years of experience relevant experience in roads and highways construction |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Next Review Date</b>                                   | 17/01/2030                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>NSQF Approval Date</b>                                 | 17/01/2025                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



## Qualification Pack

|                       |                             |
|-----------------------|-----------------------------|
| Reference code on NQR | QG-04-CO-03456-2025-V1-TICE |
| NQR Version           | 1                           |



## Qualification Pack

# ICE/CON/N5001: Conduct construction material testing using the Universal Testing Machine (UTM)

## Description

This unit covers the skills and knowledge required to perform tensile, compressive, flexural, and shear strength tests using a UTM to ensure material compliance with industry standards.

## Scope

The scope covers the following :

- Set up the UTM for use
- Prepare the sample
- Conduct tests on samples
- Analyze the test results
- Prepare the test results report

## Elements and Performance Criteria

### *Set up the UTM for use*

To be competent, the user/individual on the job must be able to:

- PC1.** set up the UTM according to the test requirements, ensuring it is calibrated correctly.
- PC2.** select and install the appropriate fixtures or grips based on the type of test, i.e. tensile, compressive, flexural, or shear
- PC3.** secure the material with no obstructions around the machine.

### *Prepare the sample*

To be competent, the user/individual on the job must be able to:

- PC4.** select a representative sample of the construction material to be tested
- PC5.** prepare the test specimens according to the dimensions and shape specified in the relevant standards
- PC6.** inspect the sample for any defects or irregularities that could affect the test results
- PC7.** condition the specimen to ensure it's at a specific temperature or humidity before testing, as required
- PC8.** ensure the samples are free from defects and within the tolerance limits for the specific test

### *Conduct tests on samples*

To be competent, the user/individual on the job must be able to:

- PC9.** perform tensile test by placing the sample in the tensile grips and gradually applying the load
- PC10.** maintain the record of elongation and ultimate tensile strength
- PC11.** perform compressive test by positioning the sample between the compression plates and applying the load
- PC12.** measure the compressive strength and deformation during the compressive test



## Qualification Pack

- PC13.** conduct the flexural test by supporting the sample on two points and applying the load at the center
- PC14.** record the flexural strength and deflection
- PC15.** conduct the shear test by securing the sample and applying the load perpendicular to the axis to measure the shear strength
- PC16.** use the UTM's software or data acquisition system to capture the load, displacement, and stress-strain data accurately
- PC17.** monitor the test progress, ensuring it aligns with the expected results

### Analyze the test results

To be competent, the user/individual on the job must be able to:

- PC18.** analyze the collected data to determine the material's mechanical properties, such as yield strength, ultimate strength, modulus of elasticity, and shear modulus
- PC19.** compare the results against the industry standards to assess material compliance
- PC20.** identify any anomalies or deviations in the test results and determine possible causes, such as machine calibration issues, sample defects, or incorrect test execution

### Prepare the test results report

To be competent, the user/individual on the job must be able to:

- PC21.** prepare the test report including the methodology, test conditions, observed data, and analysis results
- PC22.** ensure the report is clear, concise, and follows the format required by industry standards or organizational protocols

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different materials, such as steel, concrete, and composite materials commonly used in construction
- KU2.** The mechanical properties such as tensile strength, compressive strength, elasticity, and ductility of different construction materials
- KU3.** Different components of a Universal Testing Machine (UTM) such as load cells, grips, crosshead, and control panels
- KU4.** The calibration procedures for a UTM to ensure accurate measurements
- KU5.** The safety protocols to be followed when operating the UTM to prevent accidents
- KU6.** The process of conducting tensile testing using a UTM to determine the tensile strength and elongation of a material
- KU7.** The process of conducting compressive testing using a UTM to determine the compressive strength of a material
- KU8.** The process of conducting bending tests using a UTM to measure the flexural strength of material
- KU9.** The process of conducting shear testing to measure a material's resistance to shear forces
- KU10.** How to interpret stress-strain curves and other graphical data generated by the UTM
- KU11.** How to analyze test results from a UTM to determine if the material meets the required standards and specifications



## Qualification Pack

- KU12.** The relevant industry standards, such as ASTM and ISO, that govern construction material testing
- KU13.** The importance of adhering to the industry standards and testing procedures for material testing
- KU14.** The importance and process of documenting material testing procedures, observations, and results accurately
- KU15.** The importance of recording data systematically for future reference and quality control
- KU16.** How to identify and troubleshoot common issues with the UTM, such as misalignment or load cell errors
- KU17.** The importance of following appropriate corrective actions during troubleshooting to ensure the accuracy and reliability of test results

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** Read, interpret, and write simple instructions, reports, and documentation
- GS3.** Perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** Work effectively in a team, supporting colleagues and contributing to team goals
- GS5.** Assist others, share knowledge, and seek help, when necessary
- GS6.** Manage time effectively and prioritize tasks to meet deadlines
- GS7.** Adapt to changing work environments, tasks, and processes
- GS8.** Focus on tasks and ensure accuracy in work
- GS9.** Assess situations, consider potential solutions, and implement the best course of action





## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Set up the UTM for use</i>                                                                                                             | <b>4</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC1.</b> set up the UTM according to the test requirements, ensuring it is calibrated correctly.                                       | -            | -               | -             | -          |
| <b>PC2.</b> select and install the appropriate fixtures or grips based on the type of test, i.e. tensile, compressive, flexural, or shear | -            | -               | -             | -          |
| <b>PC3.</b> secure the material with no obstructions around the machine.                                                                  | -            | -               | -             | -          |
| <i>Prepare the sample</i>                                                                                                                 | <b>7</b>     | <b>10</b>       | -             | <b>5</b>   |
| <b>PC4.</b> select a representative sample of the construction material to be tested                                                      | -            | -               | -             | -          |
| <b>PC5.</b> prepare the test specimens according to the dimensions and shape specified in the relevant standards                          | -            | -               | -             | -          |
| <b>PC6.</b> inspect the sample for any defects or irregularities that could affect the test results                                       | -            | -               | -             | -          |
| <b>PC7.</b> condition the specimen to ensure it's at a specific temperature or humidity before testing, as required                       | -            | -               | -             | -          |
| <b>PC8.</b> ensure the samples are free from defects and within the tolerance limits for the specific test                                | -            | -               | -             | -          |
| <i>Conduct tests on samples</i>                                                                                                           | <b>12</b>    | <b>24</b>       | -             | <b>9</b>   |
| <b>PC9.</b> perform tensile test by placing the sample in the tensile grips and gradually applying the load                               | -            | -               | -             | -          |
| <b>PC10.</b> maintain the record of elongation and ultimate tensile strength                                                              | -            | -               | -             | -          |
| <b>PC11.</b> perform compressive test by positioning the sample between the compression plates and applying the load                      | -            | -               | -             | -          |
| <b>PC12.</b> measure the compressive strength and deformation during the compressive test                                                 | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> conduct the flexural test by supporting the sample on two points and applying the load at the center                                                                    | -            | -               | -             | -          |
| <b>PC14.</b> record the flexural strength and deflection                                                                                                                             | -            | -               | -             | -          |
| <b>PC15.</b> conduct the shear test by securing the sample and applying the load perpendicular to the axis to measure the shear strength                                             | -            | -               | -             | -          |
| <b>PC16.</b> use the UTM's software or data acquisition system to capture the load, displacement, and stress-strain data accurately                                                  | -            | -               | -             | -          |
| <b>PC17.</b> monitor the test progress, ensuring it aligns with the expected results                                                                                                 | -            | -               | -             | -          |
| <i>Analyze the test results</i>                                                                                                                                                      | <b>4</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC18.</b> analyze the collected data to determine the material's mechanical properties, such as yield strength, ultimate strength, modulus of elasticity, and shear modulus       | -            | -               | -             | -          |
| <b>PC19.</b> compare the results against the industry standards to assess material compliance                                                                                        | -            | -               | -             | -          |
| <b>PC20.</b> identify any anomalies or deviations in the test results and determine possible causes, such as machine calibration issues, sample defects, or incorrect test execution | -            | -               | -             | -          |
| <i>Prepare the test results report</i>                                                                                                                                               | <b>3</b>     | <b>4</b>        | -             | <b>1</b>   |
| <b>PC21.</b> prepare the test report including the methodology, test conditions, observed data, and analysis results                                                                 | -            | -               | -             | -          |
| <b>PC22.</b> ensure the report is clear, concise, and follows the format required by industry standards or organizational protocols                                                  | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                     | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                 |
|----------------------------|---------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5001                                                                   |
| <b>NOS Name</b>            | Conduct construction material testing using the Universal Testing Machine (UTM) |
| <b>Sector</b>              | Construction                                                                    |
| <b>Sub-Sector</b>          |                                                                                 |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                           |
| <b>NSQF Level</b>          | 4                                                                               |
| <b>Credits</b>             | 2                                                                               |
| <b>Version</b>             | 1.0                                                                             |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                      |
| <b>Next Review Date</b>    | 17/01/2030                                                                      |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                      |



## Qualification Pack

### ICE/CON/N5002: Conduct Ground Penetrating Radar (GPR) testing

#### Description

This unit covers the use of non-invasive radar technology to detect and analyze subsurface features in construction materials, enabling the identification of structures, voids, and potential hazards without the need for excavation or drilling.

#### Scope

The scope covers the following :

- Plan and prepare for GPR testing
- Conduct survey design
- Collect the data
- Interpret the data
- Prepare the test reports
- Maintain the GPR equipment

#### Elements and Performance Criteria

##### *Plan and prepare for GPR testing*

To be competent, the user/individual on the job must be able to:

- PC1.** Conduct a preliminary site assessment to identify the specific areas for subsurface evaluation
- PC2.** Calibrate the GPR equipment according to the manufacturer's guidelines, considering the specific site conditions, such as soil type, moisture content, and expected depth of investigation
- PC3.** Set up barriers and signage to protect the GPR equipment

##### *Conduct survey design*

To be competent, the user/individual on the job must be able to:

- PC4.** Establish a grid or linear layout for the GPR survey, depending on the area being assessed
- PC5.** Mark the grid points clearly on the ground to ensure consistent data collection across the site
- PC6.** Adjust the GPR settings such as antenna frequency, scan rate, and depth range based on the material type and the level of detail required
- PC7.** use higher frequencies for better resolution with less depth penetration, and lower frequencies for deeper penetration and less detail
- PC8.** follow up with additional tests, as required, based on the GPR findings

##### *Collect the data*

To be competent, the user/individual on the job must be able to:

- PC9.** conduct the survey by moving the GPR unit systematically along the pre-marked grid lines or paths
- PC10.** monitor the GPR data in real-time on the display unit, looking for anomalies or irregular reflections that might indicate subsurface features such as voids, cracks, or changes in material composition



## Qualification Pack

- PC11.** ensure all the collected data is accurately logged with corresponding grid locations for later analysis, e.g. linking GPS data to each scan line for precise geo-referencing

### *Interpret the data*

To be competent, the user/individual on the job must be able to:

- PC12.** analyze the recorded GPR data to identify subsurface features, identifying different reflection patterns produced by different materials
- PC13.** create 3D models of the subsurface conditions, as required to get a more comprehensive view of the infrastructure's internal state

### *Prepare the test reports*

To be competent, the user/individual on the job must be able to:

- PC14.** generate reports including GPR profiles, interpreted data, identified anomalies and recommendations based on the findings, such as areas that may require further investigation or repair
- PC15.** maintain a record of all GPR surveys conducted, including raw data, calibration settings, and final reports for quality assurance and future reference

### *Maintain the GPR equipment*

To be competent, the user/individual on the job must be able to:

- PC16.** clean and inspect the GPR equipment after the survey, ensuring it is in good working condition for future use
- PC17.** calibrate the equipment periodically to maintain accuracy

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principles of GPR, including the use of electromagnetic waves to detect and map subsurface features
- KU2.** how electromagnetic waves travel through different materials and the impact of material properties on wave behavior
- KU3.** the relevant standards and guidelines for GPR testing in construction, such as ASTM or ISO standards
- KU4.** the legal and regulatory requirements for conducting GPR surveys on construction sites
- KU5.** different applications of GPR in construction
- KU6.** different components of a GPR system, including the control unit, antenna, and data storage device
- KU7.** how to calibrate the GPR equipment for accurate measurements
- KU8.** the appropriate antenna frequency for different materials and depths
- KU9.** how to prepare the testing site, including surface conditions and environmental factors that may affect the GPR readings
- KU10.** how to design a survey grid for systematic data collection
- KU11.** the techniques for scanning surfaces, maintaining consistent speed, and ensuring comprehensive coverage
- KU12.** the dielectric properties of construction materials and how they affect GPR signal reflection
- KU13.** how materials like concrete, asphalt, soil, and rebar interact with GPR signals



## Qualification Pack

- KU14.** how to interpret GPR data, including identifying anomalies, layers, and embedded objects
- KU15.** the use of GPR software for data processing, visualization, and analysis
- KU16.** the methods for estimating the depth of features detected by GPR
- KU17.** the safety precautions to protect against the electromagnetic radiation emitted by GPR equipment
- KU18.** the safety procedures for working on construction sites
- KU19.** the appropriate methods to verify the accuracy of GPR data, such as using reference markers or cross-referencing with other testing methods
- KU20.** how to document GPR testing procedures, data, and findings in a clear and organized manner

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** Read, interpret, and write simple instructions, reports, and documentation
- GS3.** Perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** Use basic IT tools and applications for effective data management
- GS5.** Work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** Assist others, share knowledge, and seek help, when necessary
- GS7.** Manage time effectively and prioritize tasks to meet deadlines
- GS8.** Adapt to changing work environments, tasks, and processes



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Plan and prepare for GPR testing</i>                                                                                                                                                                              | <b>7</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC1.</b> Conduct a preliminary site assessment to identify the specific areas for subsurface evaluation                                                                                                           | -            | -               | -             | -          |
| <b>PC2.</b> Calibrate the GPR equipment according to the manufacturer's guidelines, considering the specific site conditions, such as soil type, moisture content, and expected depth of investigation               | -            | -               | -             | -          |
| <b>PC3.</b> Set up barriers and signage to protect the GPR equipment                                                                                                                                                 | -            | -               | -             | -          |
| <i>Conduct survey design</i>                                                                                                                                                                                         | <b>9</b>     | <b>15</b>       | -             | <b>6</b>   |
| <b>PC4.</b> Establish a grid or linear layout for the GPR survey, depending on the area being assessed                                                                                                               | -            | -               | -             | -          |
| <b>PC5.</b> Mark the grid points clearly on the ground to ensure consistent data collection across the site                                                                                                          | -            | -               | -             | -          |
| <b>PC6.</b> Adjust the GPR settings such as antenna frequency, scan rate, and depth range based on the material type and the level of detail required                                                                | -            | -               | -             | -          |
| <b>PC7.</b> use higher frequencies for better resolution with less depth penetration, and lower frequencies for deeper penetration and less detail                                                                   | -            | -               | -             | -          |
| <b>PC8.</b> follow up with additional tests, as required, based on the GPR findings                                                                                                                                  | -            | -               | -             | -          |
| <i>Collect the data</i>                                                                                                                                                                                              | <b>5</b>     | <b>9</b>        | -             | <b>4</b>   |
| <b>PC9.</b> conduct the survey by moving the GPR unit systematically along the pre-marked grid lines or paths                                                                                                        | -            | -               | -             | -          |
| <b>PC10.</b> monitor the GPR data in real-time on the display unit, looking for anomalies or irregular reflections that might indicate subsurface features such as voids, cracks, or changes in material composition | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> ensure all the collected data is accurately logged with corresponding grid locations for later analysis, e.g. linking GPS data to each scan line for precise geo-referencing              | -            | -               | -             | -          |
| <i>Interpret the data</i>                                                                                                                                                                              | <b>3</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC12.</b> analyze the recorded GPR data to identify subsurface features, identifying different reflection patterns produced by different materials                                                  | -            | -               | -             | -          |
| <b>PC13.</b> create 3D models of the subsurface conditions, as required to get a more comprehensive view of the infrastructure's internal state                                                        | -            | -               | -             | -          |
| <i>Prepare the test reports</i>                                                                                                                                                                        | <b>3</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC14.</b> generate reports including GPR profiles, interpreted data, identified anomalies and recommendations based on the findings, such as areas that may require further investigation or repair | -            | -               | -             | -          |
| <b>PC15.</b> maintain a record of all GPR surveys conducted, including raw data, calibration settings, and final reports for quality assurance and future reference                                    | -            | -               | -             | -          |
| <i>Maintain the GPR equipment</i>                                                                                                                                                                      | <b>3</b>     | <b>4</b>        | -             | <b>2</b>   |
| <b>PC16.</b> clean and inspect the GPR equipment after the survey, ensuring it is in good working condition for future use                                                                             | -            | -               | -             | -          |
| <b>PC17.</b> calibrate the equipment periodically to maintain accuracy                                                                                                                                 | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                       | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5002                                  |
| <b>NOS Name</b>            | Conduct Ground Penetrating Radar (GPR) testing |
| <b>Sector</b>              | Construction                                   |
| <b>Sub-Sector</b>          |                                                |
| <b>Occupation</b>          | Quality Assurance and Quality Control          |
| <b>NSQF Level</b>          | 4                                              |
| <b>Credits</b>             | 2                                              |
| <b>Version</b>             | 1.0                                            |
| <b>Last Reviewed Date</b>  | 17/01/2025                                     |
| <b>Next Review Date</b>    | 17/01/2030                                     |
| <b>NSQC Clearance Date</b> | 17/01/2025                                     |



## Qualification Pack

### ICE/CON/N5003: Carry out Digital Concrete Testing

#### Description

This unit covers the use of advanced digital tools and sensors to assess the strength, durability, and overall quality of concrete, ensuring compliance with construction standards and optimizing material performance.

#### Scope

The scope covers the following :

- Prepare the sample
- Set up the equipment
- Conduct the test
- Conduct data analysis
- Maintain the equipment

#### Elements and Performance Criteria

##### *Prepare the sample*

To be competent, the user/individual on the job must be able to:

- PC1.** collect concrete samples from the construction site according to standard guidelines
- PC2.** form the samples into cubes or cylinders, as required
- PC3.** cure the samples in controlled conditions to ensure they represent the actual performance of the concrete in the structure

##### *Set up the equipment*

To be competent, the user/individual on the job must be able to:

- PC4.** calibrate the Digital Concrete Testing Machine (DCTM) to ensure accuracy, verifying the load cell, setting up the testing parameter
- PC5.** perform appropriate checks to ensure DCTM is in proper working condition
- PC6.** set up the digital interface by inputting the necessary test parameters such as sample size, load rate, and test type (e.g., compressive strength)

##### *Conduct the test*

To be competent, the user/individual on the job must be able to:

- PC7.** position the cured concrete sample in the machine's loading frame, ensuring it is centered and aligned to avoid uneven stress distribution
- PC8.** apply a load to the sample at a controlled rate using DCTM
- PC9.** monitor the applied load and deformation data during the test
- PC10.** check the peak load at which the concrete sample fails to determine the compressive strength of the concrete
- PC11.** follow the applicable safety guidelines, including the use of personal protective equipment (PPE) such as safety goggles, gloves, and hard hats

##### *Conduct data analysis*

To be competent, the user/individual on the job must be able to:



## Qualification Pack

- PC12.** check the machine's digital system to ensure the relevant data is recorded, e.g. load applied, displacement, and stress-strain relationships
- PC13.** analyze the data to calculate the compressive strength of the concrete
- PC14.** generate a report using the digital system, including the test results, graphs, and any observations

### *Maintain the equipment*

To be competent, the user/individual on the job must be able to:

- PC15.** clean and inspect the equipment for wear and tear
- PC16.** lubricate moving parts and check for any mechanical issues that could affect performance
- PC17.** recalibrate the equipment using traceable standards to ensure continued accuracy and maintain compliance with industry norms
- PC18.** maintain records of calibration certificates and maintenance logs

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how compressive strength defines the load-bearing capacity of concrete, including factors that influence strength like curing time and mix design
- KU2.** the tensile and flexural strength properties to assess the concrete's ability to withstand bending and tension forces
- KU3.** the effect of water permeability on the long-term performance of concrete structures
- KU4.** the proper sample preparation techniques, including curing, casting, and storing concrete samples to ensure accurate testing
- KU5.** the standard test methods like ASTM, BS, or IS codes that govern the procedures for various digital concrete tests
- KU6.** the use of different digital concrete testing equipment, such as Ultrasonic Pulse Velocity (UPV) Testers, Rebound Hammer (Schmidt Hammer) and Digital Maturity Meter
- KU7.** the importance of calibrating digital testing devices to ensure they provide accurate and reliable measurements
- KU8.** how to identify and rectify issues that may arise during testing, such as equipment malfunctions or unexpected test results
- KU9.** how to acquire, process, and interpret data from digital testing equipment, using specialized software for analysis and reporting
- KU10.** how to interpret the results obtained from digital tests, such as identification of potential issues like low strength, inconsistencies, or defects in the concrete
- KU11.** how digital test results correlate with the physical properties of concrete, helping to make informed decisions about its quality and suitability for construction
- KU12.** the safety procedures for handling and operating digital testing equipment
- KU13.** the regulatory standards and codes to be followed during concrete testing to ensure compliance with industry requirements
- KU14.** how to document the testing process, including recording data, test conditions, and any observations



## Qualification Pack

**KU15.** how to prepare detailed reports that summarize the findings, including graphical representations and analysis of the concrete's performance

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare the sample</i>                                                                                                                                      | <b>5</b>     | <b>8</b>        | -             | <b>3</b>   |
| <b>PC1.</b> collect concrete samples from the construction site according to standard guidelines                                                               | -            | -               | -             | -          |
| <b>PC2.</b> form the samples into cubes or cylinders, as required                                                                                              | -            | -               | -             | -          |
| <b>PC3.</b> cure the samples in controlled conditions to ensure they represent the actual performance of the concrete in the structure                         | -            | -               | -             | -          |
| <i>Set up the equipment</i>                                                                                                                                    | <b>6</b>     | <b>8</b>        | -             | <b>4</b>   |
| <b>PC4.</b> calibrate the Digital Concrete Testing Machine (DCTM) to ensure accuracy, verifying the load cell, setting up the testing parameter                | -            | -               | -             | -          |
| <b>PC5.</b> perform appropriate checks to ensure DCTM is in proper working condition                                                                           | -            | -               | -             | -          |
| <b>PC6.</b> set up the digital interface by inputting the necessary test parameters such as sample size, load rate, and test type (e.g., compressive strength) | -            | -               | -             | -          |
| <i>Conduct the test</i>                                                                                                                                        | <b>9</b>     | <b>15</b>       | -             | <b>6</b>   |
| <b>PC7.</b> position the cured concrete sample in the machine's loading frame, ensuring it is centered and aligned to avoid uneven stress distribution         | -            | -               | -             | -          |
| <b>PC8.</b> apply a load to the sample at a controlled rate using DCTM                                                                                         | -            | -               | -             | -          |
| <b>PC9.</b> monitor the applied load and deformation data during the test                                                                                      | -            | -               | -             | -          |
| <b>PC10.</b> check the peak load at which the concrete sample fails to determine the compressive strength of the concrete                                      | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> follow the applicable safety guidelines, including the use of personal protective equipment (PPE) such as safety goggles, gloves, and hard hats | -            | -               | -             | -          |
| <i>Conduct data analysis</i>                                                                                                                                 | <b>4</b>     | <b>9</b>        | -             | <b>3</b>   |
| <b>PC12.</b> check the machine's digital system to ensure the relevant data is recorded, e.g. load applied, displacement, and stress-strain relationships    | -            | -               | -             | -          |
| <b>PC13.</b> analyze the data to calculate the compressive strength of the concrete                                                                          | -            | -               | -             | -          |
| <b>PC14.</b> generate a report using the digital system, including the test results, graphs, and any observations                                            | -            | -               | -             | -          |
| <i>Maintain the equipment</i>                                                                                                                                | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC15.</b> clean and inspect the equipment for wear and tear                                                                                               | -            | -               | -             | -          |
| <b>PC16.</b> lubricate moving parts and check for any mechanical issues that could affect performance                                                        | -            | -               | -             | -          |
| <b>PC17.</b> recalibrate the equipment using traceable standards to ensure continued accuracy and maintain compliance with industry norms                    | -            | -               | -             | -          |
| <b>PC18.</b> maintain records of calibration certificates and maintenance logs                                                                               | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                             | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                       |
|----------------------------|---------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5003                         |
| <b>NOS Name</b>            | Carry out Digital Concrete Testing    |
| <b>Sector</b>              | Construction                          |
| <b>Sub-Sector</b>          |                                       |
| <b>Occupation</b>          | Quality Assurance and Quality Control |
| <b>NSQF Level</b>          | 4                                     |
| <b>Credits</b>             | 2                                     |
| <b>Version</b>             | 1.0                                   |
| <b>Last Reviewed Date</b>  | 17/01/2025                            |
| <b>Next Review Date</b>    | 17/01/2030                            |
| <b>NSQC Clearance Date</b> | 17/01/2025                            |



## Qualification Pack

# ICE/CON/N5004: Carry out Ultrasonic Testing (UT) of construction materials

## Description

This unit covers the use of high-frequency sound waves to evaluate the integrity, thickness, and internal structure of construction materials, enabling the detection of defects such as cracks, voids, and inclusions without causing damage to the material.

## Scope

The scope covers the following :

- Set up the UT equipment for use
- Prepare the surface
- Conduct ultrasonic testing
- Analyze the signal for flaw detection
- Document and report test results
- Carry out post-test procedures

## Elements and Performance Criteria

### *Set up the UT equipment for use*

To be competent, the user/individual on the job must be able to:

- PC1.** select the appropriate ultrasonic transducer and UT device based on the material type, thickness, and the type of flaw expected
- PC2.** use a calibration block with known dimensions and properties similar to the test material
- PC3.** calibrate the UT device by adjusting the gain, range, and other settings to match the known characteristics of the calibration block

### *Prepare the surface*

To be competent, the user/individual on the job must be able to:

- PC4.** clean the surface of the material to ensure proper contact between the transducer and the material
- PC5.** apply the appropriate couplant, e.g. gel and oil, to the surface of the material to facilitate the transmission of ultrasonic waves between the transducer and the material

### *Conduct ultrasonic testing*

To be competent, the user/individual on the job must be able to:

- PC6.** place the transducer on the material's surface, ensuring firmness and even pressure
- PC7.** generate ultrasonic waves using the transducer to travel through the material being tested
- PC8.** scan the entire area of interest by moving the transducer to ensure complete coverage to detect all possible flaws

### *Analyze the signal for flaw detection*

To be competent, the user/individual on the job must be able to:

- PC9.** analyze the reflected signals (A-scan, B-scan, or C-scan) on the UT device's screen





## Qualification Pack

- PC10.** identify the location, size, and nature of flaws based on the time of flight, amplitude, and shape of the signals
- PC11.** compare the detected signals with the reference standards to determine if the flaws are within acceptable limits
- PC12.** assess whether the material meets the required quality standards or if further action is needed, such as repair or rejection of the material

### *Document and report test results*

To be competent, the user/individual on the job must be able to:

- PC13.** record the results, including the location, size, and type of any detected flaws, including the calibration data, equipment settings, and material specifications also
- PC14.** prepare a detailed report summarizing the findings, including any recommendations for further action

### *Carry out post-test procedures*

To be competent, the user/individual on the job must be able to:

- PC15.** clean the transducer and UT device
- PC16.** dispose of the couplant and other materials used during the test following the applicable regulations

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principles of ultrasonic testing, including how high-frequency sound waves are used to detect internal flaws, measure thickness, and characterize materials
- KU2.** how ultrasonic waves propagate through different materials and how they are affected by interfaces, defects, and material boundaries
- KU3.** different applications of ultrasonic testing in construction
- KU4.** the ultrasonic testing equipment, including transducers, probes, and couplants, as well as the function of each component in the testing process
- KU5.** how to calibrate ultrasonic testing equipment using reference standards to ensure accurate readings
- KU6.** the digital display, data acquisition, and analysis software used with ultrasonic testing equipment
- KU7.** how to prepare the surface of construction materials to ensure proper contact and signal transmission, including the application of couplants
- KU8.** various scan techniques, such as A-scan, B-scan, and C-scan, and when to use each method depending on the material and type of test
- KU9.** how to interpret the echoes or signals received during ultrasonic testing to identify defects, discontinuities, or material thickness
- KU10.** the acoustic properties of different construction materials and how these properties affect ultrasonic wave behaviour
- KU11.** the common defects in construction materials, such as cracks, voids, inclusions, and delaminations, and how they appear in ultrasonic test results
- KU12.** how to analyze the ultrasonic signals, including amplitude, frequency, and time of flight, to determine the presence, size, and location of defects



## Qualification Pack

- KU13.** how to measure the thickness of materials using ultrasonic testing and how to interpret the results to assess material integrity
- KU14.** the use of appropriate software tools to process and visualize UT data
- KU15.** the safety protocols for operating ultrasonic testing equipment, including proper handling of probes and protection against potential hazards
- KU16.** the relevant industry standards and codes for ultrasonic testing, such as ASTM, ISO, or ASME, and how to apply them in practice
- KU17.** how to accurately record test results, including documenting the testing conditions, equipment settings, and observations
- KU18.** how to generate detailed reports that summarize the findings of the ultrasonic test, including visual aids such as graphs or cross-sectional images
- KU19.** the importance and process of verifying and validating the accuracy of ultrasonic testing results through comparison with other testing methods or by using known standards

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Set up the UT equipment for use</i>                                                                                                                                                  | <b>6</b>     | <b>9</b>        | -             | <b>3</b>   |
| <b>PC1.</b> select the appropriate ultrasonic transducer and UT device based on the material type, thickness, and the type of flaw expected                                             | -            | -               | -             | -          |
| <b>PC2.</b> use a calibration block with known dimensions and properties similar to the test material                                                                                   | -            | -               | -             | -          |
| <b>PC3.</b> calibrate the UT device by adjusting the gain, range, and other settings to match the known characteristics of the calibration block                                        | -            | -               | -             | -          |
| <i>Prepare the surface</i>                                                                                                                                                              | <b>4</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC4.</b> clean the surface of the material to ensure proper contact between the transducer and the material                                                                          | -            | -               | -             | -          |
| <b>PC5.</b> apply the appropriate couplant, e.g. gel and oil, to the surface of the material to facilitate the transmission of ultrasonic waves between the transducer and the material | -            | -               | -             | -          |
| <i>Conduct ultrasonic testing</i>                                                                                                                                                       | <b>6</b>     | <b>10</b>       | -             | <b>5</b>   |
| <b>PC6.</b> place the transducer on the material's surface, ensuring firmness and even pressure                                                                                         | -            | -               | -             | -          |
| <b>PC7.</b> generate ultrasonic waves using the transducer to travel through the material being tested                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> scan the entire area of interest by moving the transducer to ensure complete coverage to detect all possible flaws                                                          | -            | -               | -             | -          |
| <i>Analyze the signal for flaw detection</i>                                                                                                                                            | <b>8</b>     | <b>13</b>       | -             | <b>6</b>   |
| <b>PC9.</b> analyze the reflected signals (A-scan, B-scan, or C-scan) on the UT device's screen                                                                                         | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> identify the location, size, and nature of flaws based on the time of flight, amplitude, and shape of the signals                                                      | -            | -               | -             | -          |
| <b>PC11.</b> compare the detected signals with the reference standards to determine if the flaws are within acceptable limits                                                       | -            | -               | -             | -          |
| <b>PC12.</b> assess whether the material meets the required quality standards or if further action is needed, such as repair or rejection of the material                           | -            | -               | -             | -          |
| <i>Document and report test results</i>                                                                                                                                             | <b>3</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC13.</b> record the results, including the location, size, and type of any detected flaws, including the calibration data, equipment settings, and material specifications also | -            | -               | -             | -          |
| <b>PC14.</b> prepare a detailed report summarizing the findings, including any recommendations for further action                                                                   | -            | -               | -             | -          |
| <i>Carry out post-test procedures</i>                                                                                                                                               | <b>3</b>     | <b>6</b>        | -             | <b>2</b>   |
| <b>PC15.</b> clean the transducer and UT device                                                                                                                                     | -            | -               | -             | -          |
| <b>PC16.</b> dispose of the couplant and other materials used during the test following the applicable regulations                                                                  | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                    | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5004                                               |
| <b>NOS Name</b>            | Carry out Ultrasonic Testing (UT) of construction materials |
| <b>Sector</b>              | Construction                                                |
| <b>Sub-Sector</b>          |                                                             |
| <b>Occupation</b>          | Quality Assurance and Quality Control                       |
| <b>NSQF Level</b>          | 4                                                           |
| <b>Credits</b>             | 2                                                           |
| <b>Version</b>             | 1.0                                                         |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                  |
| <b>Next Review Date</b>    | 17/01/2030                                                  |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                  |



## Qualification Pack

# ICE/CON/N5005: Carry out Eddy Current Testing (ECT) of construction materials

## Description

This unit involves employing electromagnetic induction to identify surface and near-surface flaws, evaluate material thickness, and assess conductivity, ensuring the structural integrity and quality of construction materials.

## Scope

The scope covers the following :

- Prepare for ECT
- Conduct ECT
- Document and report ECT result

## Elements and Performance Criteria

### *Prepare for ECT*

To be competent, the user/individual on the job must be able to:

- PC1.** select appropriate material for testing, ensuring it is conductive
- PC2.** clean the material surface to remove any dirt, grease, or coatings that might interfere with the eddy current signal
- PC3.** set up the eddy current testing instrument for use as per the manufacturer's guidelines
- PC4.** select the appropriate probe based on the material's properties and the type of defects being inspected
- PC5.** calibrate the instrument using reference standards or calibration blocks with known defects to ensure accuracy

### *Conduct ECT*

To be competent, the user/individual on the job must be able to:

- PC6.** position the ECT probe on the surface of the material and induce eddy currents in the conductive material
- PC7.** scan the entire surface of the material, moving the probe over the surface of the material in a systematic manner
- PC8.** maintain the probe in a perpendicular position to the surface for consistent results
- PC9.** analyze the signals displayed on the instrument, and check for variations in the signal to identify the presence of defects or inconsistencies in the material
- PC10.** determine the type, size, and location of the defect based on the signal patterns

### *Document and report ECT result*

To be competent, the user/individual on the job must be able to:

- PC11.** record the results of the inspection, including any identified defects, their locations, and severity
- PC12.** provide recommendations for repair, further inspection, or monitoring based on the findings



## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different applications of ECT in construction
- KU2.** the principles of eddy current testing
- KU3.** electromagnetic induction and how variations in material properties affect the flow of eddy currents
- KU4.** the industry standards and codes relevant to ECT, such as ASTM, ISO, or ASME, and how to apply them in testing practices
- KU5.** the use of different ECT instruments, including probes, coils, and signal processing units
- KU6.** different types of probes used in ECT, e.g. surface probes, encircling coils, bobbin probes, and their applications based on the geometry and material being tested
- KU7.** how to calibrate ECT equipment using reference standards to ensure accurate and reliable measurements
- KU8.** how material conductivity influences eddy current response, including the effect of material thickness, composition, and temperature on test results
- KU9.** the concept of magnetic permeability and its impact on eddy current testing, particularly in ferromagnetic materials
- KU10.** how surface conditions (e.g., roughness, coating, oxidation) affect the eddy current signal
- KU11.** the proper surface preparation techniques to ensure accurate contact between the probe and the material surface
- KU12.** the scanning techniques to detect surface and subsurface defects, including the appropriate speed and orientation of probe movement
- KU13.** the selection of appropriate test frequencies based on material properties and the type of defects being sought
- KU14.** the common defects detectable by ECT in construction materials, such as cracks, corrosion, wear, and inclusions
- KU15.** how to interpret ECT signals, including the amplitude changes that indicate the presence of flaws or variations in material properties
- KU16.** how to estimate the depth of flaws based on signal attenuation and phase shift
- KU17.** the process of analyzing eddy current signals using both manual and software-assisted methods to identify and characterize defects
- KU18.** how to use graphical displays, e.g. impedance plane diagrams, to visualize eddy current data and distinguish between different types of flaws
- KU19.** how to compare test results against industry standards and acceptance criteria to assess material quality
- KU20.** the safety procedures for conducting ECT, including safe handling of electrical equipment and awareness of potential hazards
- KU21.** how to accurately record test data, including test parameters, equipment settings, and observations
- KU22.** how to generate comprehensive reports detailing the testing process, results, and any identified defects, supported by graphical data representations



## Qualification Pack

- KU23.** the methods for verifying the accuracy of ECT results, such as using multiple probe frequencies or cross-referencing with other non-destructive testing (NDT) methods

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work





## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for ECT</i>                                                                                                                                                     | <b>13</b>    | <b>22</b>       | -             | <b>8</b>   |
| <b>PC1.</b> select appropriate material for testing, ensuring it is conductive                                                                                             | -            | -               | -             | -          |
| <b>PC2.</b> clean the material surface to remove any dirt, grease, or coatings that might interfere with the eddy current signal                                           | -            | -               | -             | -          |
| <b>PC3.</b> set up the eddy current testing instrument for use as per the manufacturer's guidelines                                                                        | -            | -               | -             | -          |
| <b>PC4.</b> select the appropriate probe based on the material's properties and the type of defects being inspected                                                        | -            | -               | -             | -          |
| <b>PC5.</b> calibrate the instrument using reference standards or calibration blocks with known defects to ensure accuracy                                                 | -            | -               | -             | -          |
| <i>Conduct ECT</i>                                                                                                                                                         | <b>13</b>    | <b>22</b>       | -             | <b>9</b>   |
| <b>PC6.</b> position the ECT probe on the surface of the material and induce eddy currents in the conductive material                                                      | -            | -               | -             | -          |
| <b>PC7.</b> scan the entire surface of the material, moving the probe over the surface of the material in a systematic manner                                              | -            | -               | -             | -          |
| <b>PC8.</b> maintain the probe in a perpendicular position to the surface for consistent results                                                                           | -            | -               | -             | -          |
| <b>PC9.</b> analyze the signals displayed on the instrument, and check for variations in the signal to identify the presence of defects or inconsistencies in the material | -            | -               | -             | -          |
| <b>PC10.</b> determine the type, size, and location of the defect based on the signal patterns                                                                             | -            | -               | -             | -          |
| <i>Document and report ECT result</i>                                                                                                                                      | <b>4</b>     | <b>6</b>        | -             | <b>3</b>   |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> record the results of the inspection, including any identified defects, their locations, and severity | -            | -               | -             | -          |
| <b>PC12.</b> provide recommendations for repair, further inspection, or monitoring based on the findings           | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                   | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5005                                                  |
| <b>NOS Name</b>            | Carry out Eddy Current Testing (ECT) of construction materials |
| <b>Sector</b>              | Construction                                                   |
| <b>Sub-Sector</b>          |                                                                |
| <b>Occupation</b>          | Quality Assurance and Quality Control                          |
| <b>NSQF Level</b>          | 4                                                              |
| <b>Credits</b>             | 1                                                              |
| <b>Version</b>             | 1.0                                                            |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                     |
| <b>Next Review Date</b>    | 17/01/2030                                                     |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                     |



## Qualification Pack

### ICE/CON/N5006: Conduct tests on soil, aggregate, bitumen, steel, different construction mixes, and NDT on concrete

#### Description

This unit outlines the standards for conducting various material tests on soil, aggregate, bitumen, steel, different types of construction mixes, and performing non-destructive testing (NDT) on concrete.

#### Scope

The scope covers the following :

- Conduct soil tests
- Conduct aggregate tests
- Conduct bitumen tests
- Conduct steel tests
- Test GSB, WMM, AIL, CTB, CTGSB, DBM, BC, SMA, and cement concrete mixes
- Conduct Non-Destructive Testing (NDT) on concrete

#### Elements and Performance Criteria

##### *Conduct soil tests*

To be competent, the user/individual on the job must be able to:

- PC1.** prepare soil samples as per standard testing procedures
- PC2.** conduct moisture content test, Atterberg limits test (liquid limit, plastic limit, and shrinkage limit)
- PC3.** perform Proctor compaction test and California Bearing Ratio (CBR) test
- PC4.** conduct sieve analysis and hydrometer analysis for soil classification
- PC5.** record test results accurately and interpret data in line with industry standard

##### *Conduct aggregate tests*

To be competent, the user/individual on the job must be able to:

- PC6.** conduct sieve analysis of fine and coarse aggregates
- PC7.** test for specific gravity, water absorption, and aggregate impact value
- PC8.** perform tests for flakiness index, elongation index, and angularity number
- PC9.** determine the Los Angeles abrasion value for aggregate durability
- PC10.** document test results and ensure compliance with technical specifications

##### *Conduct bitumen tests*

To be competent, the user/individual on the job must be able to:

- PC11.** perform penetration, softening point, and ductility tests on bitumen
- PC12.** test for viscosity using viscometers and determine flash and fire points
- PC13.** conduct bitumen extraction and gradation analysis of extracted aggregates
- PC14.** carry out Marshall Stability test for asphalt mixes
- PC15.** ensure correct handling of hot bitumen



## Qualification Pack

### Conduct steel tests

To be competent, the user/individual on the job must be able to:

- PC16.** conduct tensile strength test on steel bars and rods using Universal Testing Machine (UTM)
- PC17.** perform bend and rebend tests to determine the ductility of steel
- PC18.** test for chemical composition using spectrometer or chemical analysis methods
- PC19.** conduct impact test to measure toughness of steel

### Test GSB, WMM, AIL, CTB, CTGSB, DBM, BC, SMA, and cement concrete mixes

To be competent, the user/individual on the job must be able to:

- PC20.** sample and test different types of construction mixes like granular sub-base (GSB), wet mix macadam (WMM), asphaltic concrete (DBM, BC, SMA), and cement-treated bases (CTB, CTGSB)
- PC21.** conduct gradation, compaction, and moisture content tests for GSB and WMM
- PC22.** perform Marshall Stability test for DBM, BC, and SMA mixes
- PC23.** test the compressive strength of cement concrete using cubes and cylinders
- PC24.** perform air void analysis and measure mix density for asphalt mixes

### Conduct Non-Destructive Testing (NDT) on concrete

To be competent, the user/individual on the job must be able to:

- PC25.** perform Rebound Hammer test to assess surface hardness of concrete
- PC26.** conduct Ultrasonic Pulse Velocity (UPV) test to measure concrete uniformity and detect cracks
- PC27.** use Ground Penetrating Radar (GPR) to evaluate thickness and condition of concrete structures
- PC28.** perform core cutting and extraction for in-depth analysis without damaging the structure
- PC29.** document NDT results and recommend further testing or corrective measures if required

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of soils and their engineering properties
- KU2.** standard procedures for sampling and testing soil (ASTM, AASHTO)
- KU3.** use of soil testing equipment such as sieves, Proctor molds, and CBR testing machines
- KU4.** the soil classification systems (USCS, AASHTO)
- KU5.** the procedures for calculating moisture content, density, and bearing capacity
- KU6.** the characteristics of fine and coarse aggregates
- KU7.** the methods for sampling and testing aggregates, e.g. ASTM, IS standards, etc.
- KU8.** the use of equipment such as sieves, pycnometers, and abrasion testing machines
- KU9.** the significance of aggregate gradation and its impact on concrete and asphalt quality
- KU10.** the procedures for assessing the strength, durability, and suitability of aggregates for various construction applications
- KU11.** the types and grades of bitumen used in road construction
- KU12.** the standard bitumen testing procedures, e.g. ASTM, AASHTO and IS



## Qualification Pack

- KU13.** the equipment used for bitumen testing like penetrometers, ring-and-ball apparatus, and viscometers
- KU14.** the procedures for analyzing bitumen properties such as viscosity, ductility, and softening point
- KU15.** the safety procedures for working with hot bitumen
- KU16.** the types of steel used in construction, e.g. TMT and structural steel
- KU17.** the testing standards for steel, e.g. ASTM, IS and AASHTO
- KU18.** the properties of steel such as yield strength, tensile strength, and elongation
- KU19.** the use of UTM, spectrometers, and impact testing machines
- KU20.** the procedures for recording and interpreting test data
- KU21.** the composition and uses of various construction mixes (GSB, WMM, DBM, BC, SMA)
- KU22.** the standard testing procedures for asphalt and cement concrete mixes (ASTM, AASHTO)
- KU23.** the use of equipment for mix testing such as compactors, Marshall apparatus, and compression machines
- KU24.** the importance of gradation, compaction, and stability in road construction
- KU25.** the procedures for testing cement concrete, including slump tests and compressive strength tests
- KU26.** the types and purposes of non-destructive tests for concrete
- KU27.** the use of NDT equipment like rebound hammers, ultrasonic pulse velocity meters, and GPR machines
- KU28.** the procedures for interpreting results from NDT methods
- KU29.** the factors affecting concrete integrity (cracks, voids, and anomalies)
- KU30.** the safety measures to be taken during NDT, especially when working at heights or with complex structures
- KU31.** the safe practices for handling chemicals, equipment, and testing materials
- KU32.** the use of appropriate Personal Protective Equipment (PPE)
- KU33.** the applicable environmental regulations
- KU34.** the recording and interpretation of test data
- KU35.** the preparation of test reports
- KU36.** the industry standards concerning quality assurance in material testing

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary



## Qualification Pack

**GS7.** manage time effectively and prioritize tasks to meet deadlines

**GS8.** adapt to changing work environments, tasks, and processes

**GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Conduct soil tests</i>                                                                                           | <b>5</b>     | <b>9</b>        | -             | <b>3</b>   |
| <b>PC1.</b> prepare soil samples as per standard testing procedures                                                 | -            | -               | -             | -          |
| <b>PC2.</b> conduct moisture content test, Atterberg limits test (liquid limit, plastic limit, and shrinkage limit) | -            | -               | -             | -          |
| <b>PC3.</b> perform Proctor compaction test and California Bearing Ratio (CBR) test                                 | -            | -               | -             | -          |
| <b>PC4.</b> conduct sieve analysis and hydrometer analysis for soil classification                                  | -            | -               | -             | -          |
| <b>PC5.</b> record test results accurately and interpret data in line with industry standard                        | -            | -               | -             | -          |
| <i>Conduct aggregate tests</i>                                                                                      | <b>5</b>     | <b>9</b>        | -             | <b>4</b>   |
| <b>PC6.</b> conduct sieve analysis of fine and coarse aggregates                                                    | -            | -               | -             | -          |
| <b>PC7.</b> test for specific gravity, water absorption, and aggregate impact value                                 | -            | -               | -             | -          |
| <b>PC8.</b> perform tests for flakiness index, elongation index, and angularity number                              | -            | -               | -             | -          |
| <b>PC9.</b> determine the Los Angeles abrasion value for aggregate durability                                       | -            | -               | -             | -          |
| <b>PC10.</b> document test results and ensure compliance with technical specifications                              | -            | -               | -             | -          |
| <i>Conduct bitumen tests</i>                                                                                        | <b>5</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC11.</b> perform penetration, softening point, and ductility tests on bitumen                                   | -            | -               | -             | -          |
| <b>PC12.</b> test for viscosity using viscometers and determine flash and fire points                               | -            | -               | -             | -          |
| <b>PC13.</b> conduct bitumen extraction and gradation analysis of extracted aggregates                              | -            | -               | -             | -          |





## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> carry out Marshall Stability test for asphalt mixes                                                                                                                                 | -            | -               | -             | -          |
| <b>PC15.</b> ensure correct handling of hot bitumen                                                                                                                                              | -            | -               | -             | -          |
| <i>Conduct steel tests</i>                                                                                                                                                                       | <b>4</b>     | <b>7</b>        | -             | <b>3</b>   |
| <b>PC16.</b> conduct tensile strength test on steel bars and rods using Universal Testing Machine (UTM)                                                                                          | -            | -               | -             | -          |
| <b>PC17.</b> perform bend and rebend tests to determine the ductility of steel                                                                                                                   | -            | -               | -             | -          |
| <b>PC18.</b> test for chemical composition using spectrometer or chemical analysis methods                                                                                                       | -            | -               | -             | -          |
| <b>PC19.</b> conduct impact test to measure toughness of steel                                                                                                                                   | -            | -               | -             | -          |
| <i>Test GSB, WMM, AIL, CTB, CTGSB, DBM, BC, SMA, and cement concrete mixes</i>                                                                                                                   | <b>6</b>     | <b>9</b>        | -             | <b>3</b>   |
| <b>PC20.</b> sample and test different types of construction mixes like granular sub-base (GSB), wet mix macadam (WMM), asphaltic concrete (DBM, BC, SMA), and cement-treated bases (CTB, CTGSB) | -            | -               | -             | -          |
| <b>PC21.</b> conduct gradation, compaction, and moisture content tests for GSB and WMM                                                                                                           | -            | -               | -             | -          |
| <b>PC22.</b> perform Marshall Stability test for DBM, BC, and SMA mixes                                                                                                                          | -            | -               | -             | -          |
| <b>PC23.</b> test the compressive strength of cement concrete using cubes and cylinders                                                                                                          | -            | -               | -             | -          |
| <b>PC24.</b> perform air void analysis and measure mix density for asphalt mixes                                                                                                                 | -            | -               | -             | -          |
| <i>Conduct Non-Destructive Testing (NDT) on concrete</i>                                                                                                                                         | <b>5</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC25.</b> perform Rebound Hammer test to assess surface hardness of concrete                                                                                                                  | -            | -               | -             | -          |
| <b>PC26.</b> conduct Ultrasonic Pulse Velocity (UPV) test to measure concrete uniformity and detect cracks                                                                                       | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC27.</b> use Ground Penetrating Radar (GPR) to evaluate thickness and condition of concrete structures | -            | -               | -             | -          |
| <b>PC28.</b> perform core cutting and extraction for in-depth analysis without damaging the structure      | -            | -               | -             | -          |
| <b>PC29.</b> document NDT results and recommend further testing or corrective measures if required         | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                           | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                     |
|----------------------------|-----------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5006                                                                                       |
| <b>NOS Name</b>            | Conduct tests on soil, aggregate, bitumen, steel, different construction mixes, and NDT on concrete |
| <b>Sector</b>              | Construction                                                                                        |
| <b>Sub-Sector</b>          |                                                                                                     |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                                               |
| <b>NSQF Level</b>          | 4                                                                                                   |
| <b>Credits</b>             | 2                                                                                                   |
| <b>Version</b>             | 1.0                                                                                                 |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                                          |
| <b>Next Review Date</b>    | 17/01/2030                                                                                          |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                                          |



## Qualification Pack

# ICE/CON/N5007: Carry out Highway Testing through FWD, NSV, Benkelman Beam, and Total Station

## Description

This unit is about conducting various highway testing techniques using Falling Weight Deflectometer (FWD), Network Survey Vehicle (NSV), Benkelman Beam, and Total Station to assess the structural capacity, surface characteristics, and geometric accuracy of highways.

## Scope

The scope covers the following :

- Highway testing using Falling Weight Deflectometer (FWD)
- Highway surface assessment using Network Survey Vehicle (NSV)
- Highway testing using Benkelman Beam
- Highway geometric testing using Total Station

## Elements and Performance Criteria

### *Highway testing using Falling Weight Deflectometer (FWD)*

To be competent, the user/individual on the job must be able to:

- PC1.** set up the Falling Weight Deflectometer (FWD) on the test location as per standard procedures
- PC2.** calibrate the FWD before the test to ensure accurate measurements
- PC3.** conduct deflection tests by applying dynamic loads to the pavement surface and measure the resulting deflection
- PC4.** record deflection data across multiple points along the highway
- PC5.** interpret deflection data to evaluate pavement stiffness and structural capacity
- PC6.** document test results and suggest necessary maintenance or rehabilitation measures based on findings

### *Highway surface assessment using Network Survey Vehicle (NSV)*

To be competent, the user/individual on the job must be able to:

- PC7.** prepare the Network Survey Vehicle (NSV) for a highway surface assessment, ensuring all sensors and cameras are operational
- PC8.** drive the NSV along the specified route while maintaining a consistent speed to capture accurate data
- PC9.** measure surface parameters like roughness (IRI), rutting, and cracking using on-board sensors
- PC10.** use the NSV's laser profilers to generate a detailed surface profile of the highway
- PC11.** download and process data collected by the NSV for further analysis
- PC12.** prepare reports based on surface condition data and recommend maintenance or surface repair activities

### *Highway testing using Benkelman Beam*



## Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC13.** set up the Benkelman Beam at the designated test points on the highway
- PC14.** calibrate the Benkelman Beam apparatus to ensure accurate deflection measurements
- PC15.** conduct static loading tests using a standard vehicle and measure pavement rebound deflection
- PC16.** record deflection values at multiple points to assess pavement flexibility and bearing capacity
- PC17.** analyze deflection data to determine the pavement's residual life and identify areas requiring strengthening or reconstruction
- PC18.** prepare and submit detailed reports based on Benkelman Beam test results

### *Highway geometric testing using Total Station*

To be competent, the user/individual on the job must be able to:

- PC19.** set up the Total Station at pre-determined survey points along the highway
- PC20.** calibrate the Total Station and establish control points for accurate measurements
- PC21.** conduct a topographic survey to measure highway elevations, gradients, and alignments
- PC22.** record horizontal and vertical angles, distances, and coordinates of various highway features
- PC23.** use the Total Station data to verify the highway's geometric design parameters, including cross-sections and horizontal/vertical curves
- PC24.** process survey data using computer-aided design (CAD) or geographic information system (GIS) software
- PC25.** document and report any geometric inconsistencies that could impact highway safety and performance

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of pavement deflection and how FWD assesses pavement structural integrity
- KU2.** the components and operation of FWD equipment, including load cells, geophones, and data acquisition systems
- KU3.** the standard test procedures for FWD as per ASTM, AASHTO, or IRC guidelines
- KU4.** pavement layers and their response to dynamic loading
- KU5.** how to interpret the deflection data to identify weak areas and potential pavement failure points
- KU6.** the working principles of the NSV and its components, such as laser profilers, GPS, and cameras
- KU7.** the surface condition parameters measured by NSV, including International Roughness Index (IRI), rut depth, and crack mapping
- KU8.** the standard protocols for operating an NSV on highways
- KU9.** the use of appropriate data processing software to analyze NSV outputs and generate reports
- KU10.** how to interpret the surface condition data and understand its implications for highway maintenance and performance



## Qualification Pack

- KU11.** the basic principles of deflection testing and the purpose of the Benkelman Beam test in assessing pavement flexibility
- KU12.**
  - the use of Benkelman Beam equipment, including its calibration and use in static load tests
  - the testing procedures for Benkelman Beam as per IRC or ASTM standards
- KU13.** how to interpret the pavement deflection data to evaluate its current condition and potential service life
- KU14.** the procedures for correlating deflection results with pavement design and performance expectations
- KU15.** the basic principles of geometric design and the use of Total Station for surveying
- KU16.** the components of Total Station equipment and their calibration for accurate measurement
- KU17.** the techniques for measuring horizontal and vertical angles, distances, and elevations using a Total Station
- KU18.** the use of CAD and GIS software to analyze and present survey data for highway projects
- KU19.** how to interpret the survey results and identify deviations from geometric design standards

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Highway testing using Falling Weight Deflectometer (FWD)</i>                                                                         | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC1.</b> set up the Falling Weight Deflectometer (FWD) on the test location as per standard procedures                               | -            | -               | -             | -          |
| <b>PC2.</b> calibrate the FWD before the test to ensure accurate measurements                                                           | -            | -               | -             | -          |
| <b>PC3.</b> conduct deflection tests by applying dynamic loads to the pavement surface and measure the resulting deflection             | -            | -               | -             | -          |
| <b>PC4.</b> record deflection data across multiple points along the highway                                                             | -            | -               | -             | -          |
| <b>PC5.</b> interpret deflection data to evaluate pavement stiffness and structural capacity                                            | -            | -               | -             | -          |
| <b>PC6.</b> document test results and suggest necessary maintenance or rehabilitation measures based on findings                        | -            | -               | -             | -          |
| <i>Highway surface assessment using Network Survey Vehicle (NSV)</i>                                                                    | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC7.</b> prepare the Network Survey Vehicle (NSV) for a highway surface assessment, ensuring all sensors and cameras are operational | -            | -               | -             | -          |
| <b>PC8.</b> drive the NSV along the specified route while maintaining a consistent speed to capture accurate data                       | -            | -               | -             | -          |
| <b>PC9.</b> measure surface parameters like roughness (IRI), rutting, and cracking using on-board sensors                               | -            | -               | -             | -          |
| <b>PC10.</b> use the NSV's laser profilers to generate a detailed surface profile of the highway                                        | -            | -               | -             | -          |
| <b>PC11.</b> download and process data collected by the NSV for further analysis                                                        | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> prepare reports based on surface condition data and recommend maintenance or surface repair activities                                  | -            | -               | -             | -          |
| <i>Highway testing using Benkelman Beam</i>                                                                                                          | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC13.</b> set up the Benkelman Beam at the designated test points on the highway                                                                  | -            | -               | -             | -          |
| <b>PC14.</b> calibrate the Benkelman Beam apparatus to ensure accurate deflection measurements                                                       | -            | -               | -             | -          |
| <b>PC15.</b> conduct static loading tests using a standard vehicle and measure pavement rebound deflection                                           | -            | -               | -             | -          |
| <b>PC16.</b> record deflection values at multiple points to assess pavement flexibility and bearing capacity                                         | -            | -               | -             | -          |
| <b>PC17.</b> analyze deflection data to determine the pavement's residual life and identify areas requiring strengthening or reconstruction          | -            | -               | -             | -          |
| <b>PC18.</b> prepare and submit detailed reports based on Benkelman Beam test results                                                                | -            | -               | -             | -          |
| <i>Highway geometric testing using Total Station</i>                                                                                                 | <b>9</b>     | <b>14</b>       | -             | <b>5</b>   |
| <b>PC19.</b> set up the Total Station at pre-determined survey points along the highway                                                              | -            | -               | -             | -          |
| <b>PC20.</b> calibrate the Total Station and establish control points for accurate measurements                                                      | -            | -               | -             | -          |
| <b>PC21.</b> conduct a topographic survey to measure highway elevations, gradients, and alignments                                                   | -            | -               | -             | -          |
| <b>PC22.</b> record horizontal and vertical angles, distances, and coordinates of various highway features                                           | -            | -               | -             | -          |
| <b>PC23.</b> use the Total Station data to verify the highway's geometric design parameters, including cross-sections and horizontal/vertical curves | -            | -               | -             | -          |
| <b>PC24.</b> process survey data using computer-aided design (CAD) or geographic information system (GIS) software                                   | -            | -               | -             | -          |





## Qualification Pack

| Assessment Criteria for Outcomes                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> document and report any geometric inconsistencies that could impact highway safety and performance | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                               |
|----------------------------|-------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5007                                                                 |
| <b>NOS Name</b>            | Carry out Highway Testing through FWD, NSV, Benkelman Beam, and Total Station |
| <b>Sector</b>              | Construction                                                                  |
| <b>Sub-Sector</b>          |                                                                               |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                         |
| <b>NSQF Level</b>          | 4                                                                             |
| <b>Credits</b>             | 1                                                                             |
| <b>Version</b>             | 1.0                                                                           |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                    |
| <b>Next Review Date</b>    | 17/01/2030                                                                    |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                    |



## Qualification Pack

### ICE/CON/N5008: Set up the project civil lab

#### Description

This unit is about setting up a civil laboratory for construction and infrastructure projects. It covers the essentials of lab layout, equipment foundations, testing platforms, test environment chambers, and curing tanks.

#### Scope

The scope covers the following :

- Plan and design the lab layout
- Set up the equipment foundations
- Install testing platforms
- Set up test environment chambers
- Set up curing tanks

#### Elements and Performance Criteria

##### *Plan and design the lab layout*

To be competent, the user/individual on the job must be able to:

- PC1.** develop a lab layout plan considering available space, type of tests, and equipment requirements
- PC2.** ensure adequate separation of testing areas based on materials (soil, concrete, bitumen, etc.) and types of tests (destructive, non-destructive)
- PC3.** allocate space for storage of samples, materials, and chemicals
- PC4.** design the lab for efficient workflow, ensuring that sample preparation, testing, and documentation areas are logically organized
- PC5.** plan for proper ventilation, lighting, and temperature control to ensure an optimal work environment
- PC6.** ensure adherence to safety regulations and accessibility standards in the lab layout

##### *Set up the equipment foundations*

To be competent, the user/individual on the job must be able to:

- PC7.** assess the weight and operational requirements of each major piece of equipment to determine foundation specifications
- PC8.** prepare foundations for heavy equipment such as Universal Testing Machines (UTM), compression testing machines, and vibrating equipment
- PC9.** ensure that foundations are isolated from general lab flooring to prevent vibrations and ensure stability during testing
- PC10.** install reinforcement and casting for equipment foundations to ensure durability and strength
- PC11.** inspect and maintain foundations to prevent settlement or cracking over time

##### *Install testing platforms*

To be competent, the user/individual on the job must be able to:



## Qualification Pack

- PC12.** identify testing areas for specific material tests (concrete, soil, bitumen, steel) and design suitable platforms for testing equipment
- PC13.** set up vibration-resistant platforms for precision testing equipment, such as balance scales and sieves
- PC14.** construct load-bearing platforms for compression testing, tensile testing, and other high-force testing machines
- PC15.** ensure platforms are level and capable of supporting the weight and operational stresses of testing procedures
- PC16.** establish platforms for sample preparation and ensure ease of access to nearby equipment and testing areas

### *Set up test environment chambers*

To be competent, the user/individual on the job must be able to:

- PC17.** design and construct controlled-environment chambers for specialized tests, such as humidity, temperature, and environmental simulations
- PC18.** install temperature-controlled chambers for curing concrete samples and other temperature-sensitive materials
- PC19.** ensure proper insulation and climate control systems are in place to maintain constant environmental conditions within test chambers
- PC20.** set up humidity-controlled rooms or cabinets for tests that require specific moisture conditions
- PC21.** regularly calibrate and maintain chambers to ensure accurate environmental conditions are maintained during testing

### *Set up curing tanks*

To be competent, the user/individual on the job must be able to:

- PC22.** select appropriate curing tanks based on the volume and size of concrete specimens to be cured
- PC23.** set up water curing tanks with proper dimensions and capacity for the specific project requirements
- PC24.** install automated water temperature control systems to maintain the recommended curing temperature for concrete samples
- PC25.** ensure regular monitoring of water levels and temperature in the curing tanks
- PC26.** maintain a clean and stable curing environment by preventing contamination of water or samples
- PC27.** record curing durations and conditions to ensure compliance with test specifications

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of lab design, focusing on functionality
- KU2.** the spatial requirements for different types of tests and equipment
- KU3.** the regulatory requirements for civil labs
- KU4.** ergonomics and space management in laboratory environments



## Qualification Pack

- KU5.** zoning of the lab for efficient operation, including separate areas for material storage, testing, and documentation
- KU6.** the load-bearing requirements for lab equipment and the importance of stable foundations
- KU7.** the materials and techniques for constructing durable and vibration-resistant foundations
- KU8.** the specifications for isolating foundations to prevent interference with testing accuracy
- KU9.** the procedures for monitoring and maintaining foundation stability over the long term
- KU10.** the regulations and guidelines for setting up equipment in civil labs
- KU11.** the importance of level and vibration-free platforms in ensuring accurate testing results
- KU12.** the materials and construction methods for building testing platforms that meet load-bearing and stability requirements
- KU13.** the appropriate considerations when constructing and using testing platforms in the lab
- KU14.** the purpose and importance of test environment chambers for controlling external variables during material testing
- KU15.** the types of environmental chambers (temperature-controlled, humidity-controlled) and their applications in civil labs
- KU16.** the calibration and maintenance procedures for temperature and humidity control systems
- KU17.** the regulatory standards for environmental chambers used in testing construction materials
- KU18.** how temperature and humidity can affect test results and the performance of construction materials
- KU19.** the purpose of curing tanks in ensuring proper hydration and strength development of concrete specimens
- KU20.** the standard requirements for curing tanks, including temperature control and water maintenance
- KU21.** the techniques for monitoring curing conditions, including temperature and water quality
- KU22.** the industry standards for curing concrete, such as ASTM or IS codes, and the importance of consistent curing conditions for accurate test results
- KU23.** the maintenance and cleaning procedures for curing tanks to prevent contamination or deterioration

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** assist others, share knowledge, and seek help, when necessary
- GS6.** work effectively in a team, supporting colleagues and contributing to team goals
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Plan and design the lab layout</i>                                                                                                                        | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC1.</b> develop a lab layout plan considering available space, type of tests, and equipment requirements                                                 | -            | -               | -             | -          |
| <b>PC2.</b> ensure adequate separation of testing areas based on materials (soil, concrete, bitumen, etc.) and types of tests (destructive, non-destructive) | -            | -               | -             | -          |
| <b>PC3.</b> allocate space for storage of samples, materials, and chemicals                                                                                  | -            | -               | -             | -          |
| <b>PC4.</b> design the lab for efficient workflow, ensuring that sample preparation, testing, and documentation areas are logically organized                | -            | -               | -             | -          |
| <b>PC5.</b> plan for proper ventilation, lighting, and temperature control to ensure an optimal work environment                                             | -            | -               | -             | -          |
| <b>PC6.</b> ensure adherence to safety regulations and accessibility standards in the lab layout                                                             | -            | -               | -             | -          |
| <i>Set up the equipment foundations</i>                                                                                                                      | <b>5</b>     | <b>9</b>        | -             | <b>4</b>   |
| <b>PC7.</b> assess the weight and operational requirements of each major piece of equipment to determine foundation specifications                           | -            | -               | -             | -          |
| <b>PC8.</b> prepare foundations for heavy equipment such as Universal Testing Machines (UTM), compression testing machines, and vibrating equipment          | -            | -               | -             | -          |
| <b>PC9.</b> ensure that foundations are isolated from general lab flooring to prevent vibrations and ensure stability during testing                         | -            | -               | -             | -          |
| <b>PC10.</b> install reinforcement and casting for equipment foundations to ensure durability and strength                                                   | -            | -               | -             | -          |
| <b>PC11.</b> inspect and maintain foundations to prevent settlement or cracking over time                                                                    | -            | -               | -             | -          |
| <i>Install testing platforms</i>                                                                                                                             | <b>5</b>     | <b>9</b>        | -             | <b>4</b>   |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> identify testing areas for specific material tests (concrete, soil, bitumen, steel) and design suitable platforms for testing equipment  | -            | -               | -             | -          |
| <b>PC13.</b> set up vibration-resistant platforms for precision testing equipment, such as balance scales and sieves                                  | -            | -               | -             | -          |
| <b>PC14.</b> construct load-bearing platforms for compression testing, tensile testing, and other high-force testing machines                         | -            | -               | -             | -          |
| <b>PC15.</b> ensure platforms are level and capable of supporting the weight and operational stresses of testing procedures                           | -            | -               | -             | -          |
| <b>PC16.</b> establish platforms for sample preparation and ensure ease of access to nearby equipment and testing areas                               | -            | -               | -             | -          |
| <i>Set up test environment chambers</i>                                                                                                               | <b>6</b>     | <b>10</b>       | -             | <b>3</b>   |
| <b>PC17.</b> design and construct controlled-environment chambers for specialized tests, such as humidity, temperature, and environmental simulations | -            | -               | -             | -          |
| <b>PC18.</b> install temperature-controlled chambers for curing concrete samples and other temperature-sensitive materials                            | -            | -               | -             | -          |
| <b>PC19.</b> ensure proper insulation and climate control systems are in place to maintain constant environmental conditions within test chambers     | -            | -               | -             | -          |
| <b>PC20.</b> set up humidity-controlled rooms or cabinets for tests that require specific moisture conditions                                         | -            | -               | -             | -          |
| <b>PC21.</b> regularly calibrate and maintain chambers to ensure accurate environmental conditions are maintained during testing                      | -            | -               | -             | -          |
| <i>Set up curing tanks</i>                                                                                                                            | <b>7</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC22.</b> select appropriate curing tanks based on the volume and size of concrete specimens to be cured                                           | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC23.</b> set up water curing tanks with proper dimensions and capacity for the specific project requirements                     | -            | -               | -             | -          |
| <b>PC24.</b> install automated water temperature control systems to maintain the recommended curing temperature for concrete samples | -            | -               | -             | -          |
| <b>PC25.</b> ensure regular monitoring of water levels and temperature in the curing tanks                                           | -            | -               | -             | -          |
| <b>PC26.</b> maintain a clean and stable curing environment by preventing contamination of water or samples                          | -            | -               | -             | -          |
| <b>PC27.</b> record curing durations and conditions to ensure compliance with test specifications                                    | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                     | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                       |
|----------------------------|---------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5008                         |
| <b>NOS Name</b>            | Set up the project civil lab          |
| <b>Sector</b>              | Construction                          |
| <b>Sub-Sector</b>          |                                       |
| <b>Occupation</b>          | Quality Assurance and Quality Control |
| <b>NSQF Level</b>          | 4                                     |
| <b>Credits</b>             | 1                                     |
| <b>Version</b>             | 1.0                                   |
| <b>Last Reviewed Date</b>  | 17/01/2025                            |
| <b>Next Review Date</b>    | 17/01/2030                            |
| <b>NSQC Clearance Date</b> | 17/01/2025                            |



## Qualification Pack

### ICE/CON/N5009: Basics of plants and machinery (Hot Mix Plant, Batching Plant, Paver, Grader, Loader)

#### Description

This unit outlines the competencies required to understand and operate basic construction plants and machinery used in road construction and civil infrastructure projects. It covers the fundamental knowledge and basic operations of the hot mix plant, batching plant, paver, grader, and loader.

#### Scope

The scope covers the following :

- Hot mix plant operations
- Batching plant operations
- Basic operation of paver
- Basic operation of grader
- Basic operation of loader

#### Elements and Performance Criteria

##### *Hot mix plant operations*

To be competent, the user/individual on the job must be able to:

- PC1.** identify and describe the components of a hot mix plant, including aggregate feeders, dryer drum, bitumen tanks, and mixing unit
- PC2.** understand the working principle of the hot mix plant, which produces asphalt concrete by heating and mixing aggregates with bitumen
- PC3.** prepare the plant for production by ensuring proper calibration of feed controls and maintaining the required temperature settings
- PC4.** monitor the plant's performance during production, ensuring the quality of the hot mix is consistent with project specifications
- PC5.** record production data and ensure proper storage and dispatch of the hot mix material
- PC6.** conduct routine maintenance checks on the plant, including cleaning filters, inspecting burners, and checking bitumen levels

##### *Batching plant operations*

To be competent, the user/individual on the job must be able to:

- PC7.** identify and describe the key components of a batching plant, including aggregate bins, cement silos, conveyors, mixers, and control systems
- PC8.** understand the process of producing concrete by accurately batching aggregates, cement, water, and additives as per mix design
- PC9.** set up the plant for production by calibrating the batching system and checking material levels in the silos and bins
- PC10.** monitor concrete production to ensure that the correct proportions of materials are used, and the consistency of the concrete meets project specifications



## Qualification Pack

- PC11.** record batching data and monitor the output to ensure the timely delivery of ready-mix concrete to construction sites
- PC12.** perform regular maintenance on the batching plant, including checking weigh scales, mixer blades, and water dosing systems

### *Basic operation of paver*

To be competent, the user/individual on the job must be able to:

- PC13.** identify the key components of a paver, including the screed, auger, conveyor system, and operator controls
- PC14.** understand the working mechanism of the paver, which spreads, levels, and compacts asphalt or concrete mix onto the road surface
- PC15.** operate the paver safely and efficiently to ensure a uniform layer of material is laid down, following the required thickness and width specifications
- PC16.** monitor the operation of the screed and make necessary adjustments to ensure consistent mat quality and smoothness
- PC17.** coordinate with other construction personnel to ensure the continuous supply of material to the paver
- PC18.** perform daily pre-operation checks, including inspecting fluid levels, belts, and the screed condition, to avoid breakdowns

### *Basic operation of grader*

To be competent, the user/individual on the job must be able to:

- PC19.** identify and describe the key parts of a Grader, including the blade (moldboard), circle, scarifier, and operator controls
- PC20.** understand the Grader's function in leveling and smoothing the surface of a road or site, preparing it for further construction
- PC21.** operate the Grader to level and shape the sub-base or base of a road, ensuring that the required slope and gradient are achieved
- PC22.** adjust the moldboard and control the machine's speed and direction to achieve a consistent and smooth surface finish
- PC23.** collaborate with site engineers to ensure grading accuracy according to design plans and specifications
- PC24.** conduct daily inspections of the Grader, checking hydraulic systems, blade condition, and tire pressure

### *Basic operation of loader*

To be competent, the user/individual on the job must be able to:

- PC25.** identify and describe the key parts of a Loader, including the bucket, lift arms, engine, and operator controls
- PC26.** understand the Loader's role in moving and loading materials such as soil, aggregate, and debris at construction sites
- PC27.** operate the Loader to safely transport materials, ensuring correct bucket positioning and load balancing to prevent spillage
- PC28.** coordinate with other construction personnel to ensure the timely movement of materials to or from designated areas
- PC29.** perform daily operational checks, including inspecting the hydraulic systems, bucket condition, and tire wear



## Qualification Pack

- PC30.** ensure proper maintenance of the Loader, including lubrication of moving parts, engine checks, and refueling

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of pavement deflection and how FWD assesses pavement structural integrity
- KU2.** the basic understanding of the hot mix production process and the role of various components in producing asphalt concrete
- KU3.** the importance of temperature control and aggregate gradation in the quality of the hot mix
- KU4.** the safety protocols for handling hot bitumen and high-temperature equipment
- KU5.** the basic troubleshooting skills for common issues with the Hot Mix Plant
- KU6.** the maintenance procedures and schedules to keep the plant in optimal working condition
- KU7.** the principles of batching and the importance of accurate proportioning in producing high-quality concrete
- KU8.** the components and operation of a batching plant and the role each plays in the production process
- KU9.** the safety precautions for handling cement, aggregates, and chemical additives
- KU10.** the troubleshooting for common operational issues in a batching plant, such as inconsistent mix proportions or equipment malfunctions
- KU11.** the regular maintenance tasks to keep the batching plant running efficiently
- KU12.** the functions and operation of a paver in road construction, with respect to material laying and levelling
- KU13.** the key factors affecting the quality of paving, such as material temperature, screed control, and rolling patterns
- KU14.** the safety protocols for operating pavers, especially when working on active construction sites
- KU15.** the basic troubleshooting techniques to resolve common issues like uneven laying or material segregation
- KU16.** routine maintenance procedures, including screed cleaning and checking hydraulic systems
- KU17.** the principles of grading and the role of the Grader in creating a level and smooth surface for construction
- KU18.** the operating procedures and controls of the Grader for both fine and rough grading
- KU19.** the safety measures when operating Graders, including working on uneven terrain and near other heavy machinery
- KU20.** the common challenges in grading, such as achieving the correct slope and dealing with difficult soil conditions
- KU21.** the routine maintenance tasks to ensure the Grader operates efficiently, including blade sharpening and hydraulic fluid checks
- KU22.** the functions of a Loader in moving materials on construction sites, particularly in road construction and earthmoving
- KU23.** the safety protocols for operating Loaders, including load balancing, correct use of operator controls, and safe maneuvering



## Qualification Pack

- KU24.** the key operational skills for efficient loading and transportation of materials
- KU25.** the troubleshooting for common issues experienced while operating Loaders, such as overloading, material spillage, and tire wear
- KU26.** the maintenance routines, such as checking tire pressure, hydraulic fluid levels, and bucket condition to ensure long-term performance

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Hot mix plant operations</i>                                                                                                                                         | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC1.</b> identify and describe the components of a hot mix plant, including aggregate feeders, dryer drum, bitumen tanks, and mixing unit                            | -            | -               | -             | -          |
| <b>PC2.</b> understand the working principle of the hot mix plant, which produces asphalt concrete by heating and mixing aggregates with bitumen                        | -            | -               | -             | -          |
| <b>PC3.</b> prepare the plant for production by ensuring proper calibration of feed controls and maintaining the required temperature settings                          | -            | -               | -             | -          |
| <b>PC4.</b> monitor the plant's performance during production, ensuring the quality of the hot mix is consistent with project specifications                            | -            | -               | -             | -          |
| <b>PC5.</b> record production data and ensure proper storage and dispatch of the hot mix material                                                                       | -            | -               | -             | -          |
| <b>PC6.</b> conduct routine maintenance checks on the plant, including cleaning filters, inspecting burners, and checking bitumen levels                                | -            | -               | -             | -          |
| <i>Batching plant operations</i>                                                                                                                                        | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC7.</b> identify and describe the key components of a batching plant, including aggregate bins, cement silos, conveyors, mixers, and control systems                | -            | -               | -             | -          |
| <b>PC8.</b> understand the process of producing concrete by accurately batching aggregates, cement, water, and additives as per mix design                              | -            | -               | -             | -          |
| <b>PC9.</b> set up the plant for production by calibrating the batching system and checking material levels in the silos and bins                                       | -            | -               | -             | -          |
| <b>PC10.</b> monitor concrete production to ensure that the correct proportions of materials are used, and the consistency of the concrete meets project specifications | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> record batching data and monitor the output to ensure the timely delivery of ready-mix concrete to construction sites                                  | -            | -               | -             | -          |
| <b>PC12.</b> perform regular maintenance on the batching plant, including checking weigh scales, mixer blades, and water dosing systems                             | -            | -               | -             | -          |
| <i>Basic operation of paver</i>                                                                                                                                     | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC13.</b> identify the key components of a paver, including the screed, auger, conveyor system, and operator controls                                            | -            | -               | -             | -          |
| <b>PC14.</b> understand the working mechanism of the paver, which spreads, levels, and compacts asphalt or concrete mix onto the road surface                       | -            | -               | -             | -          |
| <b>PC15.</b> operate the paver safely and efficiently to ensure a uniform layer of material is laid down, following the required thickness and width specifications | -            | -               | -             | -          |
| <b>PC16.</b> monitor the operation of the screed and make necessary adjustments to ensure consistent mat quality and smoothness                                     | -            | -               | -             | -          |
| <b>PC17.</b> coordinate with other construction personnel to ensure the continuous supply of material to the paver                                                  | -            | -               | -             | -          |
| <b>PC18.</b> perform daily pre-operation checks, including inspecting fluid levels, belts, and the screed condition, to avoid breakdowns                            | -            | -               | -             | -          |
| <i>Basic operation of grader</i>                                                                                                                                    | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC19.</b> identify and describe the key parts of a Grader, including the blade (moldboard), circle, scarifier, and operator controls                             | -            | -               | -             | -          |
| <b>PC20.</b> understand the Grader's function in leveling and smoothing the surface of a road or site, preparing it for further construction                        | -            | -               | -             | -          |
| <b>PC21.</b> operate the Grader to level and shape the sub-base or base of a road, ensuring that the required slope and gradient are achieved                       | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC22.</b> adjust the moldboard and control the machine's speed and direction to achieve a consistent and smooth surface finish         | -            | -               | -             | -          |
| <b>PC23.</b> collaborate with site engineers to ensure grading accuracy according to design plans and specifications                      | -            | -               | -             | -          |
| <b>PC24.</b> conduct daily inspections of the Grader, checking hydraulic systems, blade condition, and tire pressure                      | -            | -               | -             | -          |
| <i>Basic operation of loader</i>                                                                                                          | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC25.</b> identify and describe the key parts of a Loader, including the bucket, lift arms, engine, and operator controls              | -            | -               | -             | -          |
| <b>PC26.</b> understand the Loader's role in moving and loading materials such as soil, aggregate, and debris at construction sites       | -            | -               | -             | -          |
| <b>PC27.</b> operate the Loader to safely transport materials, ensuring correct bucket positioning and load balancing to prevent spillage | -            | -               | -             | -          |
| <b>PC28.</b> coordinate with other construction personnel to ensure the timely movement of materials to or from designated areas          | -            | -               | -             | -          |
| <b>PC29.</b> perform daily operational checks, including inspecting the hydraulic systems, bucket condition, and tire wear                | -            | -               | -             | -          |
| <b>PC30.</b> ensure proper maintenance of the Loader, including lubrication of moving parts, engine checks, and refueling                 | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                          | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5009                                                                         |
| <b>NOS Name</b>            | Basics of plants and machinery (Hot Mix Plant, Batching Plant, Paver, Grader, Loader) |
| <b>Sector</b>              | Construction                                                                          |
| <b>Sub-Sector</b>          |                                                                                       |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                                 |
| <b>NSQF Level</b>          | 4                                                                                     |
| <b>Credits</b>             | 1                                                                                     |
| <b>Version</b>             | 1.0                                                                                   |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                            |
| <b>Next Review Date</b>    | 17/01/2030                                                                            |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                            |



## Qualification Pack

# ICE/CON/N5010: Basics of preparation of lab program and reporting daily progress report and monthly progress report

## Description

This unit outlines the competencies required to design and implement a laboratory testing program and report the results and progress in the form of Daily Progress Reports (DPR) and Monthly Progress Reports (MPR). It covers setting up lab schedules, conducting tests, and systematically reporting project progress in construction, infrastructure, and civil engineering projects

## Scope

The scope covers the following :

- Preparation of lab testing program
- Preparation of Daily Progress Report (DPR)
- Preparation of Monthly Progress Report (MPR)

## Elements and Performance Criteria

### *Preparation of lab testing program*

To be competent, the user/individual on the job must be able to:

- PC1.** identify project requirements, specifications, and standards for testing construction materials such as soil, aggregates, bitumen, concrete, steel, etc.
- PC2.** design a comprehensive lab testing program, including a timeline for tests such as material characterization, quality control tests, and validation of fieldwork results
- PC3.** ensure that the testing program complies with project specifications, quality standards (such as IS, ASTM, or relevant local standards), and the timelines set by the project management team
- PC4.** allocate resources, including personnel, testing equipment, and materials, to ensure the smooth execution of the lab program
- PC5.** monitor the progress of the lab tests, ensuring timely completion and reporting of results
- PC6.** maintain flexibility in the testing program to accommodate project delays or changes in the scope of work

### *Preparation of Daily Progress Report (DPR)*

To be competent, the user/individual on the job must be able to:

- PC7.** collect and organize daily test data from ongoing lab tests, including observations, results, and any issues encountered
- PC8.** ensure that the DPR includes details such as the type of test conducted, date and time, test results, materials tested, and any deviations from project specifications
- PC9.** compile a summary of the day's lab activities, highlighting significant test outcomes, issues faced, and resolutions
- PC10.** coordinate with site engineers and project managers to ensure consistency between field activities and lab progress



## Qualification Pack

- PC11.** review and verify the accuracy of all reported data before submitting the DPR to the project management team
- PC12.** submit the DPR within the specified deadline, ensuring the project team has up-to-date information on lab activities and test outcomes

### *Preparation of Monthly Progress Report (MPR)*

To be competent, the user/individual on the job must be able to:

- PC13.** compile detailed test results and observations from the laboratory testing program for the entire month, organizing them into sections for easy review
- PC14.** include a summary of key performance metrics, such as the number of tests completed, pending tests, and overall lab efficiency
- PC15.** highlight critical test outcomes that affect project decisions, including major test failures, deviations from specifications, and remedial actions taken
- PC16.** provide an analysis of trends in material quality over time, identifying patterns that may impact project performance (e.g., consistent strength increases in concrete, aggregate gradation trends)
- PC17.** coordinate with site engineers, project managers, and quality control personnel to ensure that the MPR reflects both fieldwork and lab work progress
- PC18.** ensure that the MPR is reviewed by senior lab personnel for accuracy and completeness before submission
- PC19.** submit the MPR by the specified deadline to inform the project team of overall progress and performance in relation to the testing schedule

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of setting up a laboratory testing program based on project requirements, ensuring all key material tests are included
- KU2.** the testing standards and procedures for construction materials
- KU3.** scheduling and resource management for lab personnel, equipment, and materials
- KU4.** the documentation protocols for test data, maintaining clear and detailed records of test results and observations
- KU5.** quality assurance principles to ensure accurate and reliable test results
- KU6.** structure and format of a standard DPR, including what details need to be reported daily
- KU7.** the best practices for organizing and presenting test data in a clear and concise manner
- KU8.** the coordination techniques to ensure lab progress is in sync with field activities
- KU9.** the importance of timely and accurate reporting to support project decision-making
- KU10.** the tools and software used for generating DPRs, such as spreadsheets, lab management systems, or project management tools
- KU11.** the structure and format of an MPR, including comprehensive reporting of monthly lab activities
- KU12.** data analysis methods to interpret test results over time and highlight trends or issues
- KU13.** the coordination techniques for gathering data and insights from various team members, ensuring a comprehensive MPR



## Qualification Pack

- KU14.** Key Performance Indicators (KPIs) for lab work, such as test completion rates, efficiency, and accuracy of results
- KU15.** use of appropriate software tools to generate MPRs, including project management platforms or laboratory information management systems (LIMS)

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes
- GS9.** focus on tasks and ensure accuracy in work



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Preparation of lab testing program</i>                                                                                                                                                                 | <b>10</b>    | <b>16</b>       | -             | <b>6</b>   |
| <b>PC1.</b> identify project requirements, specifications, and standards for testing construction materials such as soil, aggregates, bitumen, concrete, steel, etc.                                      | -            | -               | -             | -          |
| <b>PC2.</b> design a comprehensive lab testing program, including a timeline for tests such as material characterization, quality control tests, and validation of fieldwork results                      | -            | -               | -             | -          |
| <b>PC3.</b> ensure that the testing program complies with project specifications, quality standards (such as IS, ASTM, or relevant local standards), and the timelines set by the project management team | -            | -               | -             | -          |
| <b>PC4.</b> allocate resources, including personnel, testing equipment, and materials, to ensure the smooth execution of the lab program                                                                  | -            | -               | -             | -          |
| <b>PC5.</b> monitor the progress of the lab tests, ensuring timely completion and reporting of results                                                                                                    | -            | -               | -             | -          |
| <b>PC6.</b> maintain flexibility in the testing program to accommodate project delays or changes in the scope of work                                                                                     | -            | -               | -             | -          |
| <i>Preparation of Daily Progress Report (DPR)</i>                                                                                                                                                         | <b>10</b>    | <b>17</b>       | -             | <b>6</b>   |
| <b>PC7.</b> collect and organize daily test data from ongoing lab tests, including observations, results, and any issues encountered                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> ensure that the DPR includes details such as the type of test conducted, date and time, test results, materials tested, and any deviations from project specifications                        | -            | -               | -             | -          |
| <b>PC9.</b> compile a summary of the day's lab activities, highlighting significant test outcomes, issues faced, and resolutions                                                                          | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> coordinate with site engineers and project managers to ensure consistency between field activities and lab progress                                                                                 | -            | -               | -             | -          |
| <b>PC11.</b> review and verify the accuracy of all reported data before submitting the DPR to the project management team                                                                                        | -            | -               | -             | -          |
| <b>PC12.</b> submit the DPR within the specified deadline, ensuring the project team has up-to-date information on lab activities and test outcomes                                                              | -            | -               | -             | -          |
| <i>Preparation of Monthly Progress Report (MPR)</i>                                                                                                                                                              | <b>10</b>    | <b>17</b>       | -             | <b>8</b>   |
| <b>PC13.</b> compile detailed test results and observations from the laboratory testing program for the entire month, organizing them into sections for easy review                                              | -            | -               | -             | -          |
| <b>PC14.</b> include a summary of key performance metrics, such as the number of tests completed, pending tests, and overall lab efficiency                                                                      | -            | -               | -             | -          |
| <b>PC15.</b> highlight critical test outcomes that affect project decisions, including major test failures, deviations from specifications, and remedial actions taken                                           | -            | -               | -             | -          |
| <b>PC16.</b> provide an analysis of trends in material quality over time, identifying patterns that may impact project performance (e.g., consistent strength increases in concrete, aggregate gradation trends) | -            | -               | -             | -          |
| <b>PC17.</b> coordinate with site engineers, project managers, and quality control personnel to ensure that the MPR reflects both fieldwork and lab work progress                                                | -            | -               | -             | -          |
| <b>PC18.</b> ensure that the MPR is reviewed by senior lab personnel for accuracy and completeness before submission                                                                                             | -            | -               | -             | -          |
| <b>PC19.</b> submit the MPR by the specified deadline to inform the project team of overall progress and performance in relation to the testing schedule                                                         | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                 | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5010                                                                                        |
| <b>NOS Name</b>            | Basics of preparation of lab program and reporting daily progress report and monthly progress report |
| <b>Sector</b>              | Construction                                                                                         |
| <b>Sub-Sector</b>          |                                                                                                      |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                                                |
| <b>NSQF Level</b>          | 4                                                                                                    |
| <b>Credits</b>             | 1                                                                                                    |
| <b>Version</b>             | 1.0                                                                                                  |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                                           |
| <b>Next Review Date</b>    | 17/01/2030                                                                                           |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                                           |



## Qualification Pack

# ICE/CON/N5011: Follow health, safety, and environmental standards in construction material testing

## Description

This unit outlines the health, safety, and environmental standards required for conducting construction material testing in labs and field environments. It ensures adherence to safety protocols for equipment handling, material storage, and testing procedures while fostering a culture of safety awareness.

## Scope

The scope covers the following :

- Ensure safety during UTM testing
- Ensure safety during GPR testing
- Ensure safety during ultrasonic testing
- Ensure safety during ECT
- Ensure safety during NDT
- Follow material handling and storage protocols
- Maintain equipment safety
- Follow emergency and environmental compliance standards
- Adhere to lab safety procedures

## Elements and Performance Criteria

### *Ensure safety during UTM testing*

To be competent, the user/individual on the job must be able to:

- PC1.** verify calibration and functionality of UTM before use
- PC2.** use appropriate guards and barriers to prevent accidental contact
- PC3.** follow SOPs for tensile, compressive, flexural, and shear tests, ensuring load limits are not exceeded

### *Ensure safety during GPR testing*

To be competent, the user/individual on the job must be able to:

- PC4.** use GPR in areas cleared of obstacles to prevent tripping
- PC5.** ensure all safety protocols are in place, especially in active construction or traffic zones
- PC6.** ensure the equipment's electromagnetic emissions meet regulatory standards

### *Ensure safety during ultrasonic testing*

To be competent, the user/individual on the job must be able to:

- PC7.** use couplants as per guidelines and avoid spillage to prevent slips
- PC8.** inspect transducers and ensure proper insulation of cables

### *Ensure safety during ECT*

To be competent, the user/individual on the job must be able to:

- PC9.** calibrate ECT probes to avoid inaccuracies
- PC10.** follow precautions to handle high-frequency equipment safely





## Qualification Pack

### *Ensure safety during NDT*

To be competent, the user/individual on the job must be able to:

**PC11.** adhere to safety standards for rebound hammers, pulse velocity meters, and core drills

**PC12.** ensure operators are trained to detect equipment malfunctions during use

### *Follow material handling and storage protocols*

To be competent, the user/individual on the job must be able to:

**PC13.** identify hazards associated with materials like aggregates, bitumen, and chemicals

**PC14.** store chemicals in labeled, ventilated cabinets, segregated by reactivity type

**PC15.** handle and transport materials using appropriate lifting techniques to prevent strain injuries

**PC16.** dispose of material waste, including hazardous chemicals, as per environmental regulations

### *Maintain equipment safety*

To be competent, the user/individual on the job must be able to:

**PC17.** perform regular maintenance and inspections of testing machines to ensure their safe operation

**PC18.** use lockout/tagout procedures to deactivate faulty machines during repairs

**PC19.** prevent electrical hazards by ensuring grounded connections for powered equipment

### *Follow emergency and environmental compliance standards*

To be competent, the user/individual on the job must be able to:

**PC20.** familiarize with fire safety protocols and emergency exits in the lab

**PC21.** ensure availability and proper use of first aid kits, fire extinguishers, and eyewash stations

**PC22.** participate in periodic drills for fire, chemical spills, and evacuation scenarios

**PC23.** follow spill control measures and waste segregation systems for sustainability

### *Adhere to lab safety procedures*

To be competent, the user/individual on the job must be able to:

**PC24.** ensure testing stations are clean, organized, and free of clutter

**PC25.** use fume hoods during chemical tests to prevent inhalation of harmful vapors

**PC26.** label and store samples securely to prevent contamination or spillage

**PC27.** maintain proper ventilation and lighting in the lab to ensure a conducive working environment

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** the applicable health and safety regulations (e.g., OSHA, ISO 45001)

**KU2.** the safety procedures for working on construction sites

**KU3.** the use of relevant Personal Protective Equipment (PPE) such as safety goggles, gloves, and hard hats

**KU4.** the safe handling techniques for chemicals and construction materials

**KU5.** the first aid measures for injuries related to burns, cuts, or chemical exposure

**KU6.** the environmental regulations for waste disposal and energy-efficient operations

**KU7.** the operating principles and hazards of testing machines such as UTM, GPR, and ECT



## Qualification Pack

- KU8.** the safety protocols to be followed when operating the UTM to prevent accidents
- KU9.** the safety precautions to protect against the electromagnetic radiation emitted by GPR equipment
- KU10.** the safety procedures for handling and operating digital testing equipment
- KU11.** the safety protocols for operating ultrasonic testing equipment, including proper handling of probes and protection against potential hazards
- KU12.** the safety procedures for conducting ECT, including safe handling of electrical equipment and awareness of potential hazards
- KU13.** the safety procedures for working with hot bitumen
- KU14.** the safety measures to be taken during NDT, especially when working at heights or with complex structures
- KU15.** the importance of focusing on functionality, workflow optimization, safety, ventilation, and fire safety in lab operations
- KU16.** the calibration requirements and safety features for mechanical and NDT tools
- KU17.** the preventive maintenance practices for lab and field equipment
- KU18.** the characteristics of materials such as bitumen, steel, and aggregates
- KU19.** the chemical reactivity and compatibility for safe storage and handling
- KU20.** the fire suppression techniques for electrical and chemical fires
- KU21.** the evacuation protocols for lab and field settings
- KU22.** the ergonomics for workstation setup and material lifting
- KU23.** the best practices for sample preservation and testing accuracy
- KU24.** the safety protocols for handling hot bitumen and high-temperature equipment
- KU25.** the safety precautions for handling cement, aggregates, and chemical additives
- KU26.** the safety protocols for operating pavers, especially when working on active construction sites
- KU27.** the safety measures when operating Graders, including working on uneven terrain and near other heavy machinery
- KU28.** the safety protocols for operating Loaders, including load balancing, correct use of operator controls, and safe maneuvering

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow verbal instructions, convey information clearly and concisely to team members and supervisors
- GS2.** read, interpret, and write simple instructions, reports, and documentation
- GS3.** perform basic arithmetic operations such as addition, subtraction, multiplication, and division
- GS4.** use basic IT tools and applications for effective data management
- GS5.** work effectively in a team, supporting colleagues and contributing to team goals
- GS6.** assist others, share knowledge, and seek help, when necessary
- GS7.** manage time effectively and prioritize tasks to meet deadlines
- GS8.** adapt to changing work environments, tasks, and processes



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Ensure safety during UTM testing</i>                                                                            | <b>4</b>     | <b>8</b>        | -             | <b>2</b>   |
| <b>PC1.</b> verify calibration and functionality of UTM before use                                                 | -            | -               | -             | -          |
| <b>PC2.</b> use appropriate guards and barriers to prevent accidental contact                                      | -            | -               | -             | -          |
| <b>PC3.</b> follow SOPs for tensile, compressive, flexural, and shear tests, ensuring load limits are not exceeded | -            | -               | -             | -          |
| <i>Ensure safety during GPR testing</i>                                                                            | <b>4</b>     | <b>8</b>        | -             | <b>2</b>   |
| <b>PC4.</b> use GPR in areas cleared of obstacles to prevent tripping                                              | -            | -               | -             | -          |
| <b>PC5.</b> ensure all safety protocols are in place, especially in active construction or traffic zones           | -            | -               | -             | -          |
| <b>PC6.</b> ensure the equipment's electromagnetic emissions meet regulatory standards                             | -            | -               | -             | -          |
| <i>Ensure safety during ultrasonic testing</i>                                                                     | <b>2</b>     | <b>5</b>        | -             | <b>2</b>   |
| <b>PC7.</b> use couplants as per guidelines and avoid spillage to prevent slips                                    | -            | -               | -             | -          |
| <b>PC8.</b> inspect transducers and ensure proper insulation of cables                                             | -            | -               | -             | -          |
| <i>Ensure safety during ECT</i>                                                                                    | <b>2</b>     | <b>5</b>        | -             | <b>1</b>   |
| <b>PC9.</b> calibrate ECT probes to avoid inaccuracies                                                             | -            | -               | -             | -          |
| <b>PC10.</b> follow precautions to handle high-frequency equipment safely                                          | -            | -               | -             | -          |
| <i>Ensure safety during NDT</i>                                                                                    | <b>2</b>     | <b>5</b>        | -             | <b>1</b>   |
| <b>PC11.</b> adhere to safety standards for rebound hammers, pulse velocity meters, and core drills                | -            | -               | -             | -          |
| <b>PC12.</b> ensure operators are trained to detect equipment malfunctions during use                              | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Follow material handling and storage protocols</i>                                                       | <b>4</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC13.</b> identify hazards associated with materials like aggregates, bitumen, and chemicals             | -            | -               | -             | -          |
| <b>PC14.</b> store chemicals in labeled, ventilated cabinets, segregated by reactivity type                 | -            | -               | -             | -          |
| <b>PC15.</b> handle and transport materials using appropriate lifting techniques to prevent strain injuries | -            | -               | -             | -          |
| <b>PC16.</b> dispose of material waste, including hazardous chemicals, as per environmental regulations     | -            | -               | -             | -          |
| <i>Maintain equipment safety</i>                                                                            | <b>3</b>     | <b>4</b>        | -             | <b>3</b>   |
| <b>PC17.</b> perform regular maintenance and inspections of testing machines to ensure their safe operation | -            | -               | -             | -          |
| <b>PC18.</b> use lockout/tagout procedures to deactivate faulty machines during repairs                     | -            | -               | -             | -          |
| <b>PC19.</b> prevent electrical hazards by ensuring grounded connections for powered equipment              | -            | -               | -             | -          |
| <i>Follow emergency and environmental compliance standards</i>                                              | <b>4</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC20.</b> familiarize with fire safety protocols and emergency exits in the lab                          | -            | -               | -             | -          |
| <b>PC21.</b> ensure availability and proper use of first aid kits, fire extinguishers, and eyewash stations | -            | -               | -             | -          |
| <b>PC22.</b> participate in periodic drills for fire, chemical spills, and evacuation scenarios             | -            | -               | -             | -          |
| <b>PC23.</b> follow spill control measures and waste segregation systems for sustainability                 | -            | -               | -             | -          |
| <i>Adhere to lab safety procedures</i>                                                                      | <b>5</b>     | <b>3</b>        | -             | <b>3</b>   |
| <b>PC24.</b> ensure testing stations are clean, organized, and free of clutter                              | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> use fume hoods during chemical tests to prevent inhalation of harmful vapors                  | -            | -               | -             | -          |
| <b>PC26.</b> label and store samples securely to prevent contamination or spillage                         | -            | -               | -             | -          |
| <b>PC27.</b> maintain proper ventilation and lighting in the lab to ensure a conducive working environment | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                           | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ICE/CON/N5011                                                                       |
| <b>NOS Name</b>            | Follow health, safety, and environmental standards in construction material testing |
| <b>Sector</b>              | Construction                                                                        |
| <b>Sub-Sector</b>          |                                                                                     |
| <b>Occupation</b>          | Quality Assurance and Quality Control                                               |
| <b>NSQF Level</b>          | 4                                                                                   |
| <b>Credits</b>             | 1                                                                                   |
| <b>Version</b>             | 1.0                                                                                 |
| <b>Last Reviewed Date</b>  | 17/01/2025                                                                          |
| <b>Next Review Date</b>    | 17/01/2030                                                                          |
| <b>NSQC Clearance Date</b> | 17/01/2025                                                                          |



## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:



## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.





## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings



## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Awarding Body. Each Performance Criteria (PC)/ Element will be assigned marks proportional to its importance in NOS. AB will also lay down proportion of marks for Theory and Skills Practical for each PC/ Element.
2. The assessment for the knowledge part will be based on knowledge bank of questions created by Assessment Bodies subject to approval by AB
3. Individual assessment agencies will create unique question papers for knowledge/theory part for assessment of candidates as per assessment criteria given below
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on assessment criteria.
5. The passing percentage for each QP will be 70%. To pass the Qualification Pack, every trainee should score a minimum of 70% individually in each NOS.
6. The Assessor shall check the final outcome of the practices while evaluating the steps performed to achieve the final outcome.



## Qualification Pack

7. The trainee shall be provided with a chance to repeat the test to correct his procedures in case of improper performance, with a deduction of marks for each iteration.
8. After the certain number of iterations as decided by AB the trainee is marked as fail, scoring zero marks for the procedure for the practical activity.
9. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack within the specified time frame set by AB.
10. Minimum duration of Assessment of each QP shall be of 4hrs/trainee

### Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

### Compulsory NOS

| National Occupational Standards                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| ICE/CON/N5001. Conduct construction material testing using the Universal Testing Machine (UTM) | 30           | 50              | -             | 20         | 100         | 10        |
| ICE/CON/N5002. Conduct Ground Penetrating Radar (GPR) testing                                  | 30           | 50              | -             | 20         | 100         | 8         |
| ICE/CON/N5003. Carry out Digital Concrete Testing                                              | 30           | 50              | -             | 20         | 100         | 7         |
| ICE/CON/N5004. Carry out Ultrasonic Testing (UT) of construction materials                     | 30           | 50              | -             | 20         | 100         | 8         |
| ICE/CON/N5005. Carry out Eddy Current Testing (ECT) of construction materials                  | 30           | 50              | -             | 20         | 100         | 7         |



### Qualification Pack

| National Occupational Standards                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|---------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ICE/CON/N5006. Conduct tests on soil, aggregate, bitumen, steel, different construction mixes, and NDT on concrete  | 30           | 50              | -             | 20         | 100         | 10         |
| ICE/CON/N5007. Carry out Highway Testing through FWD, NSV, Benkelman Beam, and Total Station                        | 30           | 50              | -             | 20         | 100         | 8          |
| ICE/CON/N5008. Set up the project civil lab                                                                         | 30           | 50              | -             | 20         | 100         | 10         |
| ICE/CON/N5009. Basics of plants and machinery (Hot Mix Plant, Batching Plant, Paver, Grader, Loader)                | 30           | 50              | -             | 20         | 100         | 10         |
| ICE/CON/N5010. Basics of preparation of lab program and reporting daily progress report and monthly progress report | 30           | 50              | -             | 20         | 100         | 10         |
| ICE/CON/N5011. Follow health, safety, and environmental standards in construction material testing                  | 30           | 50              | -             | 20         | 100         | 7          |
| DGT/VSQ/N0102. Employability Skills (60 Hours)                                                                      | 20           | 30              | -             | -          | 50          | 5          |
| <b>Total</b>                                                                                                        | <b>350</b>   | <b>580</b>      | <b>-</b>      | <b>220</b> | <b>1150</b> | <b>100</b> |





## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |



## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |



## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |